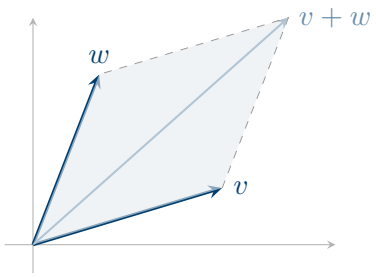
Technische Universität Hohenfeld
Faculty of Mathematics · Institute of Algebra and Geometry

Linear Algebra I

Lecture Notes, Exercises and Past Examinations

Module MA-B-101 · 9 ECTS



Prof. Dr. Henrik Lindqvist
with exercises compiled by Dr. Mira Okonkwo and the tutor team

Winter Term 2026/27
Version 1.3 of 27 October 2026 – this script will be updated during the semester

About this script. These notes accompany the lecture *Linear Algebra I* (MA-B-101) at TU Hohenfeld. They are *not* a substitute for attending the lectures: proofs are occasionally shortened, and many motivating pictures and heuristics are only given at the blackboard. Conversely, some details that are skipped in the lecture for lack of time are written out here.

The notes are based on lectures given by the author in the winter terms 2022/23, 2024/25 and 2025/26. Earlier versions were typeset from handwritten notes by Jonas Feldhaus and Ana Ribeiro, to whom I am very grateful. Thanks also go to all students who reported misprints; a list of corrections is maintained on the course page.

Version history.

v1.0	12 Oct 2026	Chapters 0–2, course information
v1.1	19 Oct 2026	Chapter 3; typos in Section 1.3 corrected (thanks to L. Brandt)
v1.2	23 Oct 2026	Chapters 4–6 (preliminary), exercise collection
v1.3	27 Oct 2026	Past exams WS 2025/26 added; Example 4.3.6 corrected

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Typeset with LaTeX. Cover illustration: the parallelogram spanned by two vectors, cf. Chapter 6.

Course Information

Welcome

Linear algebra is the study of vector spaces and of the structure-preserving maps between them. It is, together with analysis, one of the two pillars on which almost all of modern mathematics rests, and it is equally indispensable in physics, computer science, economics and engineering. In this course you will meet, perhaps for the first time, mathematics as a deductive science: every statement will be *proved* from a small number of definitions. Learning to read, understand and eventually write such proofs is at least as important a goal of this course as the material itself. Expect this to be hard at first – it is hard for everybody.

Organisation

Lecturer	Prof. Dr. Henrik Lindqvist Institute of Algebra and Geometry, Building M3, Room 2.17 `henrik.lindqvist@math.tu-hohenfeld.de` Office hours: Wednesdays 13:30–14:30 and by appointment
Coordinator	Dr. Mira Okonkwo, Room 2.21 `la1-coordination@math.tu-hohenfeld.de` Office hours: Tuesdays 10:00–11:00
Tutors	Felix Achterberg, Leonie Cramer, Daniel Esposito, Hannah Grünwald, Tomasz Kowalczyk, Selin Yıldız
Lectures	Monday 10:15–11:45, Lecture Hall H 1 (Gaede-Bau) Thursday 08:15–09:45, Lecture Hall H 1 (Gaede-Bau)
Plenary session	Friday 12:15–13:45, Lecture Hall H 4 (weekly, starting 23 October)
Tutorials	14 groups of at most 25 students, Tuesday–Thursday (see enrolment system)
Lecture period	19 October 2026 – 12 February 2027 no lectures 21 Dec 2026 – 6 Jan 2027 (winter break)
Course page	Learning platform Ilias, course “MA-B-101 Lineare Algebra I (WS 26/27)” enrolment key announced in the first lecture
Discussion forum	Ilias forum (moderated by the tutors)

Exercise sheets and admission to the exam

1. A new exercise sheet is published every **Thursday after the lecture**. Solutions are to be handed in by the following **Thursday, 08:00**, as a single PDF file via Ilias. Late submissions are not accepted, regardless of the reason; instead, your worst sheet will be discarded (see below).
2. Solutions may be handed in by teams of **at most two** students from the same tutorial group. Both names and matriculation numbers must appear on the first page. Copying solutions

from other teams, from the internet or from AI systems is a violation of the examination regulations and leads to the loss of all points on the respective sheet for all involved parties; repeated violations lead to exclusion from the course.

3. Each sheet has four problems worth 10 points each. There will be 13 sheets; the sheet with the lowest score is not counted.

4. **Admission to the final exam** ("Prüfungsvorleistung") requires

   - at least **50 %** of the total points on the 12 counted sheets, and
   - presenting a solution at the blackboard in your tutorial at least **twice**.

5. Students who obtained the admission in an earlier year keep it; they need not register for a tutorial again, but are very welcome to.

Examination

The module is concluded by a written examination of **120 minutes**. No aids are allowed except one sheet of A4 paper (both sides) with handwritten notes. Calculators, phones and smartwatches are not permitted.

	Date	**Registration deadline**
Final exam	Thursday, 18 February 2027, 09:00–11:00	31 January 2027
Resit exam	Wednesday, 7 April 2027, 14:00–16:00	21 March 2027

Registration takes place via the campus management system; registration in Ilias is *not* an exam registration. Rooms will be announced one week before the exam. Bonus points from the exercise sheets are *not* awarded. Two past exams with full solutions are included at the end of these notes (Part III); they give a realistic impression of length and difficulty.

Grading scheme (out of 60 points; subject to minor adjustment):

Points	≥ 54	≥ 51	≥ 48	≥ 45	≥ 42	≥ 39	≥ 36	≥ 33	≥ 30	< 30
Grade	1.0	1.3	1.7	2.0	2.3	2.7	3.0	3.3	3.7/4.0	5.0

Tentative schedule

Week	Dates	Topics
1	19.10. / 22.10.	Organisation; logic, sets, maps (Chapter 1)
2	26.10. / 29.10.	Relations, induction; groups (Chapters 1 and 2)
3	02.11. / 05.11.	Rings and fields, $\mathbb{Z}/n\mathbb{Z}$, complex numbers (Chapter 2)
4	09.11. / 12.11.	Linear systems, Gaussian elimination (Chapter 3)
5	16.11. / 19.11.	Vector spaces, subspaces, linear combinations (Chapter 4)
6	23.11. / 26.11.	Linear independence, bases, Steinitz exchange (Chapter 4)
7	30.11. / 03.12.	Dimension, sums of subspaces (Chapter 4)
8	07.12. / 10.12.	Linear maps, kernel, image, rank–nullity (Chapter 5)
9	14.12. / 17.12.	Matrices of linear maps, matrix multiplication (Chapter 5)
–	21.12. – 06.01.	*Winter break*
10	07.01. / 11.01.	Change of basis, rank of matrices, quotient spaces (Chapter 5)
11	14.01. / 18.01.	Permutations, determinants (Chapter 6)
12	21.01. / 25.01.	Laplace expansion, Cramer's rule (Chapter 6)
13	28.01. / 01.02.	Eigenvalues, characteristic polynomial (Chapter 7)
14	04.02. / 08.02.	Diagonalisability (Chapter 7)
15	11.02.	Question session, exam preparation

Literature

The lecture does not follow a single textbook. The following books are recommended; all are available in the library (several also as e-books via the campus network).

[1] S. Axler, *Linear Algebra Done Right*, 4th ed., Springer, 2024. *(Elegant, determinant-free approach; excellent for Chapters 3, 4 and 6.)*

[2] S. Bosch, *Lineare Algebra*, 6th ed., Springer Spektrum, 2021.

[3] G. Fischer, B. Springborn, *Lineare Algebra*, 19th ed., Springer Spektrum, 2020. *(The classical German reference.)*

[4] K. Jänich, *Lineare Algebra*, 11th ed., Springer, 2008. *(Very readable, many pictures.)*

[5] P. D. Lax, *Linear Algebra and Its Applications*, 2nd ed., Wiley, 2007.

[6] G. Strang, *Introduction to Linear Algebra*, 6th ed., Wellesley–Cambridge Press, 2023. *(Computational viewpoint; good for intuition, less so for proofs.)*

A few words of advice

- *Rework every lecture within two days.* Mathematics at university is not learned by listening, but by thinking through definitions and proofs yourself with pen and paper. Plan roughly 10–12 hours per week for this course in addition to the lectures and tutorials.
- *Know the definitions by heart.* Most failed proofs in exams fail at the very first step, because the precise definition of the notion in question is not known.
- *Work on the exercise sheets yourself first* and only then discuss them with your partner. A sheet that was copied, even partly, is worthless as preparation for the exam.
- *Ask questions* – in the lecture, in the tutorials, in the forum and in office hours. There are no stupid questions in a first-year course.

Notation and conventions

$\mathbb{N} = \{0, 1, 2, \dots\}$	natural numbers (*including* 0); $\mathbb{N}_{\geq 1} = \{1, 2, 3, \dots\}$
$\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$	integers, rational, real and complex numbers
K	always denotes a field (Definition 2.2.1); if nothing else is said, the reader may think of $K = \mathbb{R}$
$A \subseteq B$	A is a subset of B (equality allowed); $A \subsetneq B$ for proper subsets
$:=$	"is defined to be equal to"
$[n]$	the set $\{1, \dots, n\}$ for $n \in \mathbb{N}_{\geq 1}$
$K^{m \times n}$	the set of $m \times n$ matrices with entries in K
$e_1, \dots, e_n$	the standard basis vectors of K^n
■	end of a proof

Vectors in K^n are written as columns; to save space we sometimes write $(x_1, \dots, x_n)^\mathsf{T}$. Statements marked with an asterisk (*) are not relevant for the exam.

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Part I

Lecture Notes

Chapter 1

Preliminaries: Logic, Sets and Maps

This chapter fixes language and notation. Most of it will be familiar in some form from school; what may be new is the precision with which we use it. We work in "naive" set theory and do not discuss axiomatic foundations.

1.1 Statements and proofs

A *statement* (or proposition) is a sentence that is either true or false. From statements A, B we form new ones: $\neg A$ ("not A"), $A \wedge B$ ("A and B"), $A \vee B$ ("A or B", non-exclusive), $A \Rightarrow B$ ("A implies B") and $A \Leftrightarrow B$ ("A if and only if B"). Their meaning is defined by truth tables; we only recall that $A \Rightarrow B$ is false precisely when A is true and B is false. In particular, a false statement implies anything (*ex falso quodlibet*).

Lemma 1.1.1. *For statements A, B the following are* tautologies*, i.e. true independently of the truth values of A and B:*

(i) $(A \Rightarrow B) \Leftrightarrow (\neg B \Rightarrow \neg A)$ *(contraposition)*

(ii) $\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$ *and* $\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$ *(De Morgan)*

(iii) $(A \Leftrightarrow B) \Leftrightarrow \big((A \Rightarrow B) \wedge (B \Rightarrow A)\big)$

Proof. Compare truth tables; for (i) both sides are false exactly when A is true and B is false. ∎

If $A(x)$ is a statement depending on an element x of a set M, we write $\forall x \in M : A(x)$ ("for all x in M, $A(x)$ holds") and $\exists x \in M : A(x)$ ("there exists $x \in M$ with $A(x)$"); $\exists!$ means "there exists exactly one". Negation interchanges the quantifiers:

$$\neg\big(\forall x \in M : A(x)\big) \iff \exists x \in M : \neg A(x), \qquad \neg\big(\exists x \in M : A(x)\big) \iff \forall x \in M : \neg A(x).$$

The order of quantifiers matters: $\forall x\, \exists y : x < y$ is true in $\mathbb{N}$, whereas $\exists y\, \forall x : x < y$ is false.

Remark 1.1.2 (Types of proof). A *direct proof* of $A \Rightarrow B$ assumes A and derives B. A *proof by contraposition* assumes $\neg B$ and derives $\neg A$. A *proof by contradiction* assumes $\neg B$ (and A) and derives a false statement. To prove $A \Leftrightarrow B$ one usually proves both implications separately. To prove that $A_1, \dots, A_n$ are equivalent it suffices to prove $A_1 \Rightarrow A_2 \Rightarrow \cdots \Rightarrow A_n \Rightarrow A_1$.

Example 1.1.3. *Claim: $\sqrt{2}$ is irrational.* Suppose $\sqrt{2} = p/q$ with $p, q \in \mathbb{Z}$, $q \neq 0$, and the fraction in lowest terms. Then $p^2 = 2q^2$ is even, hence p is even (the square of an odd number is odd), say $p = 2r$. Then $4r^2 = 2q^2$, so $q^2 = 2r^2$ is even and q is even. Thus 2 divides both p and q, contradicting the choice of p/q in lowest terms.

1.2 Sets

A *set* is a collection of well-distinguished objects, its *elements*; we write $x \in M$ if x is an element of M. Two sets are equal if they have the same elements. Sets are described by listing their elements, $M = \{1, 2, 3\}$, or by a property, $M = \{\, x \in \mathbb{Z} \mid x^2 < 10 \,\}$. The *empty set* $\emptyset$ has no elements. For sets A, B we define

$$A \cup B := \{\, x \mid x \in A \vee x \in B \,\}, \quad A \cap B := \{\, x \mid x \in A \wedge x \in B \,\}, \quad A \setminus B := \{\, x \in A \mid x \notin B \,\}.$$

A and B are *disjoint* if $A \cap B = \emptyset$. The *power set* $\mathcal{P}(M)$ is the set of all subsets of M. For a family $(A_i)_{i\in I}$ of sets, $\bigcup_{i\in I} A_i$ and $\bigcap_{i\in I} A_i$ are defined in the obvious way. The *Cartesian product* of $A_1, \dots, A_n$ is the set of n-tuples

$$A_1 \times \cdots \times A_n := \{\, (a_1, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, \dots, n \,\},$$

where $(a_1, \dots, a_n) = (b_1, \dots, b_n)$ if and only if $a_i = b_i$ for all i. We write $A^n = A \times \cdots \times A$.

To prove an equality of sets $A = B$, one proves $A \subseteq B$ and $B \subseteq A$, i.e. $x \in A \Rightarrow x \in B$ and vice versa. This simple principle ("double inclusion") will be used over and over again.

1.3 Maps

Definition 1.3.1. Let X, Y be sets. A *map* (or function) $f\colon X \to Y$ assigns to every $x \in X$ exactly one element $f(x) \in Y$. X is the *domain*, Y the *codomain* of f. Two maps are equal if they have the same domain, the same codomain and $f(x) = g(x)$ for all x. For $A \subseteq X$ and $B \subseteq Y$,

$$f(A) := \{\, f(a) \mid a \in A \,\} \subseteq Y, \qquad f^{-1}(B) := \{\, x \in X \mid f(x) \in B \,\} \subseteq X$$

are the *image* of A and the *preimage* of B. The set of all maps $X \to Y$ is denoted Y^X or $\mathrm{Map}(X, Y)$.

Definition 1.3.2. A map $f\colon X \to Y$ is called

- *injective* if $f(x) = f(x')$ implies $x = x'$ for all $x, x' \in X$;
- *surjective* if for every $y \in Y$ there is an $x \in X$ with $f(x) = y$, i.e. $f(X) = Y$;
- *bijective* if it is injective and surjective.

For maps $f\colon X \to Y$ and $g\colon Y \to Z$ the *composition* $g \circ f\colon X \to Z$ is $x \mapsto g(f(x))$. Composition is associative: $h \circ (g \circ f) = (h \circ g) \circ f$, as both sides map x to $h(g(f(x)))$. The *identity* $\mathrm{id}_X\colon X \to X$, $x \mapsto x$, satisfies $f \circ \mathrm{id}_X = f = \mathrm{id}_Y \circ f$.

Proposition 1.3.3. *Let $f\colon X \to Y$ be a map with $X \neq \emptyset$.*

(a) f is injective $\iff$ there is $g\colon Y \to X$ with $g \circ f = \mathrm{id}_X$.

(b) f is surjective $\iff$ there is $g\colon Y \to X$ with $f \circ g = \mathrm{id}_Y$.

(c) f is bijective $\iff$ there is $g\colon Y \to X$ with $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. In this case g is uniquely determined; it is called the inverse map *f^{-1}.*

Proof. (a) "$\Leftarrow$": If $f(x) = f(x')$, then $x = g(f(x)) = g(f(x')) = x'$. "$\Rightarrow$": Fix $x_0 \in X$. For $y \in f(X)$ there is a unique x with $f(x) = y$; put $g(y) := x$. For $y \notin f(X)$ put $g(y) := x_0$. Then $g(f(x)) = x$ for all x.

(b) "$\Leftarrow$": Every y equals $f(g(y)) \in f(X)$. "$\Rightarrow$": For each $y \in Y$ choose[1] some $x_y \in f^{-1}(\{y\})$ and put $g(y) := x_y$.

(c) The equivalence follows from (a) and (b), except that in "$\Rightarrow$" we need *one* g satisfying both equations: if $g \circ f = \mathrm{id}_X$ and $f \circ g' = \mathrm{id}_Y$, then $g = g \circ \mathrm{id}_Y = g \circ (f \circ g') = (g \circ f) \circ g' = g'$. The same computation shows uniqueness. ∎

Corollary 1.3.4. *Let X, Y be finite sets with $|X| = |Y|$ and $f\colon X \to Y$. Then f is injective $\iff$ f is surjective $\iff$ f is bijective.*

[1] This uses the axiom of choice, which we accept without discussion.

Proof. If f is injective, then $|f(X)| = |X| = |Y|$, so $f(X) = Y$. If f is surjective and $f(x) = f(x')$ for some $x \neq x'$, then $|f(X)| \leq |X| - 1 < |Y|$, a contradiction. ∎

We will see a striking analogue of this corollary for linear maps between finite-dimensional vector spaces of the same dimension (Corollary 5.2.2).

1.4 Relations and induction

Definition 1.4.1. A *relation* on a set M is a subset $R \subseteq M \times M$; we write $x \sim y$ instead of $(x, y) \in R$. The relation is an *equivalence relation* if it is

(E1) reflexive: $x \sim x$ for all $x \in M$;

(E2) symmetric: $x \sim y \Rightarrow y \sim x$;

(E3) transitive: $x \sim y \wedge y \sim z \Rightarrow x \sim z$.

The *equivalence class* of x is $[x] := \{\, y \in M \mid y \sim x \,\}$, and $M/\!\sim := \{\, [x] \mid x \in M \,\}$ is the *quotient set*.

Proposition 1.4.2. *If $\sim$ is an equivalence relation on M, then any two equivalence classes are either equal or disjoint, and M is the union of all equivalence classes. In other words, $M/\!\sim$ is a* partition *of M.*

Proof. Since $x \in [x]$ by (E1), the classes cover M. Suppose $z \in [x] \cap [y]$. If $w \in [x]$, then $w \sim x$, $x \sim z$ (by (E2)) and $z \sim y$, hence $w \sim y$ by (E3), i.e. $w \in [y]$. Thus $[x] \subseteq [y]$, and by symmetry $[x] = [y]$. ∎

Example 1.4.3. Let $n \in \mathbb{N}_{\geq 1}$. On $\mathbb{Z}$ define $a \equiv b \pmod{n}$ if n divides $b - a$. This is an equivalence relation ("congruence modulo n"), and $\mathbb{Z}/n\mathbb{Z} := \mathbb{Z}/\!\equiv\, = \{[0], [1], \ldots, [n-1]\}$ has exactly n elements by division with remainder. We will turn $\mathbb{Z}/n\mathbb{Z}$ into a ring in Section 2.2.

Induction. Let $A(n)$ be a statement for every $n \in \mathbb{N}$. If $A(0)$ holds (*base case*) and $A(n) \Rightarrow A(n+1)$ holds for every $n \in \mathbb{N}$ (*induction step*), then $A(n)$ holds for all $n \in \mathbb{N}$. This is an axiom about the natural numbers. In the *strong* form, the induction step may use $A(0), \ldots, A(n)$ to prove $A(n+1)$.

Example 1.4.4. For all $n \in \mathbb{N}$, $\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$. The case $n = 0$ is clear, and if the formula holds for n, then $\sum_{k=0}^{n+1} k = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$.

Chapter 2

Algebraic Structures: Groups, Rings, Fields

Linear algebra can be done over any *field*: a set in which one can add, subtract, multiply and divide subject to the usual rules. To make this precise we first introduce groups, which also play a central role in their own right (e.g. the permutation groups in Chapter 6).

2.1 Groups

Definition 2.1.1. A *group* is a pair $(G, *)$ consisting of a set G and a map $*\colon G \times G \to G$, $(a, b) \mapsto a * b$ (the *composition*), such that

(G1) $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$ (associativity)

(G2) there is $e \in G$ with $e * a = a * e = a$ for all $a \in G$ (neutral element)

(G3) for every $a \in G$ there is $a' \in G$ with $a * a' = a' * a = e$ (inverse elements)

The group is *abelian* (or commutative) if moreover $a * b = b * a$ for all $a, b \in G$.

Lemma 2.1.2. *Let $(G, *)$ be a group.*

(a) The neutral element is unique, and every $a \in G$ has a unique inverse, denoted a^{-1}.

(b) $(a^{-1})^{-1} = a$ and $(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$.

(c) (Cancellation) If $a * b = a * c$, then $b = c$; if $b * a = c * a$, then $b = c$.

(d) For $a, b \in G$ the equation $a * x = b$ has the unique solution $x = a^{-1} * b$.

Proof. (a) If $e, \tilde{e}$ are neutral, then $e = e * \tilde{e} = \tilde{e}$. If a', a'' are inverses of a, then $a' = a' * e = a' * (a * a'') = (a' * a) * a'' = e * a'' = a''$. (b) a satisfies the defining property of an inverse of a^{-1}; and $(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * a^{-1} = e$, similarly in the other order. (c) Multiply by a^{-1} from the left (resp. right). (d) $x = a^{-1} * b$ is a solution, and it is unique by (c). ∎

Abelian groups are often written additively, $(G, +)$, with neutral element 0 and inverse $-a$; otherwise we use multiplicative notation ab, 1, a^{-1}.

Examples 2.1.3. (1) $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ are abelian groups; $(\mathbb{N}, +)$ is not (no inverses). $(\mathbb{Q} \setminus \{0\}, \cdot)$ is an abelian group; $(\mathbb{Z} \setminus \{0\}, \cdot)$ is not.

(2) For a set X, the set $S(X)$ of all bijections $X \to X$ is a group under composition with neutral element id_X (Proposition 1.3.3). For $X = [n]$ we write $S_n := S([n])$, the *symmetric group* on n letters; it has $n!$ elements and is not abelian for $n \geq 3$.

(3) The set of invertible real 2×2 matrices is a non-abelian group under matrix multiplication (see Chapter 5).

Definition 2.1.4. A subset $H \subseteq G$ of a group is a *subgroup* if $e \in H$, and $a, b \in H$ implies $ab \in H$ and $a^{-1} \in H$. A map $\varphi\colon G \to G'$ between groups is a *group homomorphism* if $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in G$. Its *kernel* is $\ker \varphi := \varphi^{-1}(\{e'\})$.

Lemma 2.1.5. *Let $\varphi\colon G \to G'$ be a group homomorphism. Then $\varphi(e) = e'$, $\varphi(a^{-1}) = \varphi(a)^{-1}$, $\ker \varphi$ is a subgroup of G, and φ is injective if and only if $\ker \varphi = \{e\}$.*

Proof. $\varphi(e) = \varphi(ee) = \varphi(e)\varphi(e)$ gives $\varphi(e) = e'$ by cancellation, and then $\varphi(a)\varphi(a^{-1}) = \varphi(e) = e'$. The subgroup property of the kernel is now immediate. If $\ker\varphi = \{e\}$ and $\varphi(a) = \varphi(b)$, then $\varphi(a^{-1}b) = \varphi(a)^{-1}\varphi(b) = e'$, so $a^{-1}b = e$, i.e. $a = b$. The converse is clear since $\varphi(e) = e'$. ∎

The criterion "injective $\iff$ trivial kernel" will reappear for linear maps and is used constantly.

2.2 Rings and fields

Definition 2.2.1. A *ring* is a triple $(R, +, \cdot)$ such that $(R, +)$ is an abelian group, multiplication is associative, there is an element $1 \in R$ with $1 \cdot a = a \cdot 1 = a$ for all a, and the distributive laws

$$a(b + c) = ab + ac, \qquad (a + b)c = ac + bc$$

hold for all $a, b, c \in R$. The ring is *commutative* if $ab = ba$ for all a, b. A *field* is a commutative ring K with $1 \neq 0$ in which every $a \neq 0$ has a multiplicative inverse; equivalently, $(K \setminus \{0\}, \cdot)$ is an abelian group, denoted $K^\times$.

Lemma 2.2.2. *In every ring R we have $0 \cdot a = a \cdot 0 = 0$ and $(-a)b = -(ab) = a(-b)$. In a field, $ab = 0$ implies $a = 0$ or $b = 0$.*

Proof. $0 \cdot a = (0+0)a = 0 \cdot a + 0 \cdot a$, hence $0 \cdot a = 0$ by cancellation in $(R, +)$. Then $ab + (-a)b = (a - a)b = 0$. In a field, if $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = a^{-1} \cdot 0 = 0$. ∎

Examples 2.2.3. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields; $\mathbb{Z}$ is a commutative ring but not a field. For any commutative ring R, the polynomials $R[x]$ in one variable form a commutative ring. The 2×2 matrices over $\mathbb{R}$ form a non-commutative ring in which $ab = 0$ is possible with $a, b \neq 0$, e.g. $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)^2 = 0$.

The rings $\mathbb{Z}/n\mathbb{Z}$. On $\mathbb{Z}/n\mathbb{Z} = \{[0], \ldots, [n-1]\}$ (Example 1.4.3) we define

$$[a] + [b] := [a + b], \qquad [a] \cdot [b] := [ab].$$

We must check that this is *well defined*, i.e. independent of the chosen representatives: if $[a] = [a']$ and $[b] = [b']$, then $a' = a + kn$, $b' = b + ln$ with $k, l \in \mathbb{Z}$, so $a' + b' = a + b + (k + l)n$ and $a'b' = ab + (al + bk + kln)n$. Hence $[a' + b'] = [a + b]$ and $[a'b'] = [ab]$. All ring axioms are inherited from $\mathbb{Z}$, so $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with n elements.

Theorem 2.2.4. *$\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is a prime number.*

Proof. If $n = 1$, then $[1] = [0]$, which is not allowed in a field. If $n = ab$ with $1 < a, b < n$, then $[a], [b] \neq [0]$ but $[a][b] = [n] = [0]$, impossible in a field. Now let $n = p$ be prime and $[a] \neq [0]$, i.e. $p \nmid a$. Consider the map $\mu_a \colon \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$, $[x] \mapsto [a][x]$. It is injective: $[ax] = [ay]$ means $p \mid a(x - y)$, and since p is prime and $p \nmid a$, we get $p \mid x - y$, i.e. $[x] = [y]$. By Corollary 1.3.4, μ_a is surjective, so there is $[x]$ with $[a][x] = [1]$. ∎

For a prime p we write $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$. Usually we omit the brackets and write $\mathbb{F}_p = \{0, 1, \ldots, p-1\}$. For instance, in $\mathbb{F}_5$ we have $2 \cdot 3 = 1$, so $2^{-1} = 3$, and $4^{-1} = 4$.

Definition 2.2.5. The *characteristic* $\operatorname{char} K$ of a field K is the smallest $n \in \mathbb{N}_{\geq 1}$ with $\underbrace{1 + \cdots + 1}_{n} = 0$, or 0 if no such n exists.

One shows easily that $\operatorname{char} K$ is either 0 or a prime. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ have characteristic 0, $\mathbb{F}_p$ has characteristic p. Fields of characteristic 2 deserve occasional extra care: there $1 + 1 = 0$, so $a = -a$ for all a, and one cannot divide by 2. (See Exercise 3.)

2.3 Complex numbers

On $\mathbb{R}^2$ define $(a,b)+(c,d) := (a+c, b+d)$ and $(a,b)\cdot(c,d) := (ac-bd, ad+bc)$. A direct (tedious) verification shows that this yields a field $\mathbb{C}$, in which $(a,0)$ is identified with $a \in \mathbb{R}$. With $i := (0,1)$ we have $i^2 = -1$ and every $z \in \mathbb{C}$ can be written uniquely as $z = a + bi$ with $a = \operatorname{Re} z$, $b = \operatorname{Im} z \in \mathbb{R}$. The *conjugate* is $\bar{z} = a - bi$ and $|z| = \sqrt{z\bar{z}} = \sqrt{a^2+b^2}$. For $z \neq 0$ one has $z^{-1} = \bar{z}/|z|^2$.

Theorem 2.3.1 (Fundamental theorem of algebra)**.** *Every non-constant polynomial $f \in \mathbb{C}[x]$ has a root in $\mathbb{C}$. Consequently f factors into linear factors: $f = c\,(x-\lambda_1)\cdots(x-\lambda_n)$ with $c, \lambda_1, \dots, \lambda_n \in \mathbb{C}$.*

The proof requires analysis and is given in a later course (e.g. Complex Analysis). The second statement follows from the first by induction on the degree, using polynomial division: if $f(\lambda) = 0$, then $f = (x-\lambda)g$ with $\deg g = \deg f - 1$.

Chapter 3

Systems of Linear Equations

Throughout, K denotes a field. Solving systems of linear equations is the oldest problem of linear algebra and the origin of most of its concepts. In this chapter we solve such systems algorithmically; the structural explanation of what we observe is given in Chapters 4 and 5.

3.1 Matrices and linear systems

Definition 3.1.1. Let $m, n \in \mathbb{N}_{\geq 1}$. An $m \times n$ *matrix* over K is a rectangular array

$$A = (a_{ij})_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, \qquad a_{ij} \in K.$$

Formally, A is a map $[m] \times [n] \to K$. The set of all such matrices is $K^{m \times n}$; we identify $K^n = K^{n \times 1}$ (column vectors). For $A \in K^{m \times n}$ and $x \in K^n$ we define $Ax \in K^m$ by

$$(Ax)_i := \sum_{j=1}^{n} a_{ij} x_j \qquad (i = 1, \ldots, m).$$

A *system of m linear equations in n unknowns* is

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \qquad \text{or briefly} \qquad Ax = b, \tag{3.1}$$

with *coefficient matrix* $A = (a_{ij}) \in K^{m \times n}$ and right-hand side $b \in K^m$. The system is *homogeneous* if $b = 0$. Its *solution set* is $L(A, b) := \{\, x \in K^n \mid Ax = b \,\}$. The *augmented matrix* is $(A \mid b) \in K^{m \times (n+1)}$.

Lemma 3.1.2. *Let $A \in K^{m \times n}$, $b \in K^m$.*

(a) $A(x + y) = Ax + Ay$ and $A(\lambda x) = \lambda Ax$ for $x, y \in K^n$, $\lambda \in K$. In particular $L(A, 0)$ contains 0 and is closed under addition and scalar multiplication.

(b) If $x_0 \in L(A, b)$, then $L(A, b) = x_0 + L(A, 0) := \{\, x_0 + y \mid y \in L(A, 0) \,\}$.

Proof. (a) $(A(x+y))_i = \sum_j a_{ij}(x_j + y_j) = \sum_j a_{ij}x_j + \sum_j a_{ij}y_j$; similarly for λx. (b) If $Ay = 0$, then $A(x_0 + y) = b$. Conversely, if $Ax = b$, then $A(x - x_0) = b - b = 0$ and $x = x_0 + (x - x_0)$. ∎

In words: *the general solution of an inhomogeneous system is one particular solution plus the general solution of the associated homogeneous system.*

3.2 Elementary row operations and row echelon form

Definition 3.2.1. The *elementary row operations* on a matrix are:

(R1) interchange two rows;

(R2) multiply a row by a scalar $\lambda \in K^\times$;

(R3) add λ times row k to row i, where $i \neq k$ and $\lambda \in K$.

Lemma 3.2.2. *If $(A' \mid b')$ arises from $(A \mid b)$ by an elementary row operation, then $L(A', b') = L(A, b)$.*

Proof. Clearly every solution of the old system solves the new one. Each elementary row operation can be undone by an elementary row operation of the same type (for (R2) multiply by λ^{-1}, for (R3) add $-\lambda$ times row k). Hence the converse holds as well. ∎

Definition 3.2.3. A matrix is in *row echelon form* if it has the shape

$$\begin{pmatrix} 0\cdots0 & \boxed{*} & *\cdots & * & *\cdots & * & *\cdots \\ 0\cdots0 & 0 & 0\cdots0 & \boxed{*} & *\cdots & * & *\cdots \\ & & & & \ddots & & \\ 0\cdots0 & 0 & 0\cdots0 & 0 & 0\cdots0 & \boxed{*} & *\cdots \\ 0\cdots0 & 0 & 0\cdots0 & 0 & 0\cdots0 & 0 & 0\cdots0 \end{pmatrix},$$

i.e. there are $r \geq 0$ and column indices $j_1 < j_2 < \cdots < j_r$ such that row i $(1 \leq i \leq r)$ starts with $a_{ij_i} \neq 0$ (the *pivot*) preceded by zeros, and rows $r+1, \ldots, m$ are zero. It is in *reduced* row echelon form if moreover all pivots equal 1 and all other entries in pivot columns are 0.

Theorem 3.2.4 (Gaussian elimination)**.** *Every matrix can be transformed into reduced row echelon form by finitely many elementary row operations.*

Proof. Induction on the number m of rows; the case $A = 0$ is trivial. Let j_1 be the first column containing a non-zero entry. By (R1) we bring a row with non-zero entry in column j_1 to the top; by (R2) we make this entry 1; by (R3) we subtract suitable multiples of row 1 from all other rows so that column j_1 becomes e_1. Applying the induction hypothesis to the matrix formed by rows $2, \ldots, m$ (row operations on these rows do not affect the zeros in columns $\leq j_1$) yields a row echelon form with pivots equal to 1; finally, using (R3) once more, we clear the entries above every pivot, working from the last pivot to the first. ∎

One can show that the *reduced* row echelon form of A is uniquely determined by A; in particular the number r of non-zero rows depends only on A. We shall see in Section 5.5 that r is the *rank* of A.

3.3 Reading off the solution

Suppose $(A \mid b)$ has been transformed to $(\tilde{A} \mid \tilde{b})$ with $\tilde{A}$ in reduced row echelon form, pivots in columns $j_1 < \cdots < j_r$. The variables $x_{j_1}, \ldots, x_{j_r}$ are called *pivot variables*, the remaining $n - r$ variables are *free*.

Theorem 3.3.1. *In the situation above:*

(a) $Ax = b$ is solvable if and only if $\tilde{b}_{r+1} = \cdots = \tilde{b}_m = 0$.

(b) In this case, the free variables may be chosen arbitrarily, and each choice determines the pivot variables uniquely via $x_{j_i} = \tilde{b}_i - \sum_{k \text{ free}} \tilde{a}_{ik} x_k$. In particular, the solution is unique if and only if $r = n$.

(c) If $m < n$, the homogeneous system $Ax = 0$ has a non-zero solution.

Proof. (a) and (b) are clear from the shape of $(\tilde{A} \mid \tilde{b})$ and Lemma 3.2.2. For (c) note that $r \le m < n$, so there is at least one free variable; choosing it to be 1 gives a non-zero solution. ∎

Statement (c) – *more unknowns than equations forces non-trivial homogeneous solutions* – is the technical heart of the theory of dimension (Theorem 4.4.1).

Example 3.3.2. Over $K = \mathbb{R}$ consider

$$\begin{aligned} x_1 + 2x_2 \qquad\quad + 3x_4 &= 1 \\ 2x_1 + 4x_2 + x_3 + 8x_4 &= 4 \\ -x_1 - 2x_2 + x_3 - x_4 &= 1. \end{aligned}$$

We transform the augmented matrix ($\mathrm{II} - 2\cdot\mathrm{I}$, $\mathrm{III} + \mathrm{I}$, then $\mathrm{III} - \mathrm{II}$):

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 1 \\ 2 & 4 & 1 & 8 & 4 \\ -1 & -2 & 1 & -1 & 1 \end{array}\right) \rightsquigarrow \left(\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{array}\right) \rightsquigarrow \left(\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

This is in reduced row echelon form with $r = 2$, pivot columns $1, 3$ and free variables $x_2 = s$, $x_4 = t$. Hence

$$L(A,b) = \left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ 0 \\ -2 \\ 1 \end{pmatrix} \;\middle|\; s, t \in \mathbb{R} \right\}.$$

Had the right-hand side of the third equation been 2 instead of 1, the last row would have read $(0\ 0\ 0\ 0 \mid 1)$, and the system would be unsolvable.

Remark 3.3.3. Gaussian elimination works verbatim over every field, e.g. $\mathbb{F}_2$ or $\mathbb{F}_5$. Note however that the solution set then is a finite set: if there are $n-r$ free variables over $\mathbb{F}_p$, a solvable system has exactly p^{n-r} solutions. Numerically, over $\mathbb{R}$, Gaussian elimination is sensitive to rounding errors; strategies such as *pivoting* are discussed in the course on Numerical Mathematics.

Remark 3.3.4 (Cost). Bringing an $n \times n$ matrix into row echelon form needs about $\frac{2}{3}n^3$ arithmetic operations. In contrast, the formula of Leibniz for determinants (Chapter 6) involves $n!$ summands – for $n = 25$ this is already more than 10^{25}.

Chapter 4

Vector Spaces

4.1 Definition and examples

Definition 4.1.1. A *vector space over* K (or K-vector space) is a set V together with two maps

$$+\colon V \times V \to V,\ (v,w) \mapsto v+w, \qquad \cdot\colon K \times V \to V,\ (\lambda, v) \mapsto \lambda v,$$

called *addition* and *scalar multiplication*, such that $(V,+)$ is an abelian group and for all $\lambda, \mu \in K$, $v, w \in V$:

(V1) $(\lambda\mu)v = \lambda(\mu v)$;

(V2) $(\lambda+\mu)v = \lambda v + \mu v$ and $\lambda(v+w) = \lambda v + \lambda w$;

(V3) $1v = v$.

Elements of V are called *vectors*, elements of K *scalars*. The neutral element of $(V,+)$ is the *zero vector* 0 (or 0_V).

Lemma 4.1.2. *In every K-vector space, $0_K\, v = 0_V$, $\lambda 0_V = 0_V$, $(-1)v = -v$, and $\lambda v = 0$ implies $\lambda = 0$ or $v = 0$.*

Proof. $0v = (0+0)v = 0v + 0v$ gives $0v = 0$; similarly $\lambda 0 = \lambda(0+0)$. Then $v + (-1)v = (1-1)v = 0v = 0$. If $\lambda v = 0$ and $\lambda \neq 0$, then $v = 1v = (\lambda^{-1}\lambda)v = \lambda^{-1}(\lambda v) = \lambda^{-1}0 = 0$. ∎

Examples 4.1.3. (1) K^n with componentwise addition and scalar multiplication. More generally $K^{m\times n}$.

(2) For any set X, the set $K^X = \mathrm{Map}(X,K)$ with $(f+g)(x) := f(x) + g(x)$ and $(\lambda f)(x) := \lambda f(x)$. For $X = [n]$ this is (essentially) K^n, for $X = \mathbb{N}$ the space of sequences in K.

(3) The polynomial ring $K[x]$ is a K-vector space; so is $K[x]_{\leq d}$, the set of polynomials of degree at most d.

(4) $\mathbb{C}$ is a vector space over $\mathbb{R}$, and $\mathbb{R}$ is a vector space over $\mathbb{Q}$. More generally, if $K \subseteq L$ is a subfield, then L is a K-vector space.

(5) $C(\mathbb{R}) = \{\, f\colon \mathbb{R} \to \mathbb{R} \mid f \text{ continuous} \,\}$ and $C^\infty(\mathbb{R})$ are real vector spaces (Analysis I).

(6) The *zero space* $\{0\}$.

4.2 Subspaces

Definition 4.2.1. A subset U of a K-vector space V is a *(linear) subspace* if

(U1) $0 \in U$;

(U2) $u, u' \in U \Rightarrow u + u' \in U$;

(U3) $\lambda \in K,\ u \in U \Rightarrow \lambda u \in U$.

A subspace is itself a vector space with the restricted operations (note that $-u = (-1)u \in U$). By Lemma 3.1.2, the solution set $L(A,0)$ of a homogeneous system is a subspace of K^n, whereas $L(A,b)$ is not if $b \neq 0$ (since $0 \notin L(A,b)$).

Examples 4.2.2. (1) The subspaces of $\mathbb{R}^2$ are $\{0\}$, the lines through the origin and $\mathbb{R}^2$ itself (this follows from Theorem 4.4.6). (2) $C^\infty(\mathbb{R}) \subseteq C(\mathbb{R}) \subseteq \mathbb{R}^\mathbb{R}$ is a chain of subspaces. (3) The set of all *symmetric* matrices $A^\mathsf{T} = A$ is a subspace of $K^{n\times n}$. (4) $\{\, f \in \mathbb{R}^\mathbb{R} \mid f(0) = 1 \,\}$ is not a subspace.

Lemma 4.2.3. *The intersection $\bigcap_{i\in I} U_i$ of any family of subspaces of V is a subspace. The union $U_1 \cup U_2$ of two subspaces is a subspace if and only if $U_1 \subseteq U_2$ or $U_2 \subseteq U_1$.*

Proof. The first claim is immediate from the definition. For the second, suppose $U_1 \not\subseteq U_2$ and $U_2 \not\subseteq U_1$; choose $u_1 \in U_1 \setminus U_2$ and $u_2 \in U_2 \setminus U_1$. If $u_1 + u_2 \in U_1$, then $u_2 = (u_1 + u_2) - u_1 \in U_1$, a contradiction; likewise $u_1 + u_2 \notin U_2$. So $U_1 \cup U_2$ is not closed under addition. ∎

Definition 4.2.4. Let $v_1, \dots, v_n \in V$. A vector of the form $\lambda_1 v_1 + \cdots + \lambda_n v_n$ with $\lambda_i \in K$ is a *linear combination* of $v_1, \dots, v_n$. For a subset $S \subseteq V$, the *span* (or linear hull) of S is

$$\operatorname{span}(S) := \{\, \lambda_1 v_1 + \cdots + \lambda_n v_n \mid n \in \mathbb{N},\ v_i \in S,\ \lambda_i \in K \,\},$$

the set of all *finite* linear combinations of elements of S; by convention $\operatorname{span}(\emptyset) = \{0\}$. If $\operatorname{span}(S) = V$, then S *generates* V.

Proposition 4.2.5. $\operatorname{span}(S)$ *is the smallest subspace of V containing S, i.e. it is a subspace, contains S, and is contained in every subspace U with $S \subseteq U$. Equivalently,* $\operatorname{span}(S) = \bigcap_{U \supseteq S} U$.

Proof. The sum of two linear combinations and a scalar multiple of a linear combination are again linear combinations, and 0 is the empty linear combination; so $\operatorname{span}(S)$ is a subspace, obviously containing S. Conversely, a subspace $U \supseteq S$ contains all linear combinations of elements of S by (U2) and (U3) and induction on the number of summands. ∎

4.3 Linear independence and bases

Definition 4.3.1. A family $(v_1, \dots, v_n)$ of vectors in V is *linearly independent* if

$$\lambda_1 v_1 + \cdots + \lambda_n v_n = 0 \quad \Longrightarrow \quad \lambda_1 = \cdots = \lambda_n = 0,$$

i.e. if the zero vector can only be written as the *trivial* linear combination. Otherwise it is *linearly dependent*. An arbitrary family $(v_i)_{i\in I}$ is linearly independent if every finite subfamily is.

Remark 4.3.2. We speak of *families* rather than sets because repetitions matter: (v, v) is linearly dependent ($1 \cdot v - 1 \cdot v = 0$), whereas the set $\{v, v\} = \{v\}$ is independent if $v \neq 0$. Any family containing 0 is dependent. A single vector v is independent iff $v \neq 0$ (Lemma 4.1.2).

Lemma 4.3.3. $(v_1, \dots, v_n)$ *is linearly dependent if and only if some v_k is a linear combination of the others, i.e. $v_k \in \operatorname{span}(v_i : i \neq k)$. If $(v_1, \dots, v_{n-1})$ is independent but $(v_1, \dots, v_n)$ is dependent, then $v_n \in \operatorname{span}(v_1, \dots, v_{n-1})$.*

Proof. If $\sum_i \lambda_i v_i = 0$ with $\lambda_k \neq 0$, then $v_k = -\sum_{i\neq k} \lambda_k^{-1} \lambda_i v_i$. Conversely, $v_k = \sum_{i\neq k} \mu_i v_i$ gives the non-trivial relation $v_k - \sum_{i\neq k} \mu_i v_i = 0$. For the second statement take a non-trivial relation $\sum_{i=1}^n \lambda_i v_i = 0$; then $\lambda_n \neq 0$, since otherwise $(v_1, \dots, v_{n-1})$ would be dependent, and we may solve for v_n. ∎

Example 4.3.4. Vectors $v_1, \dots, v_n \in K^m$ are linearly independent iff the homogeneous system $Ax = 0$ with $A = (v_1 | \cdots | v_n) \in K^{m\times n}$ has only the trivial solution, since $Ax = \sum_j x_j v_j$. By Theorem 3.3.1(c), *more than m vectors in K^m are always linearly dependent.* For instance, in $\mathbb{R}^3$, $v_1 = (1, 2, 3)^\mathsf{T}$, $v_2 = (4, 5, 6)^\mathsf{T}$, $v_3 = (7, 8, 9)^\mathsf{T}$ are dependent, because $v_1 - 2v_2 + v_3 = 0$.

Definition 4.3.5. A family $\mathcal{B} = (v_i)_{i\in I}$ in V is a *basis* of V if it is linearly independent and generates V.

Examples 4.3.6. (1) The standard basis $(e_1, \dots, e_n)$ of K^n. (2) The *monomials* $(1, x, x^2, \dots)$ form a basis of $K[x]$, and $(1, x, \dots, x^d)$ is a basis of $K[x]_{\le d}$. (3) $(1, i)$ is a basis of $\mathbb{C}$ over $\mathbb{R}$. (4) The matrices E_{kl} with entry 1 at position (k, l) and 0 elsewhere form a basis of $K^{m\times n}$. (5) The empty family is a basis of $\{0\}$.

Theorem 4.3.7 (Characterisation of bases)**.** *For a family $\mathcal{B} = (v_1, \dots, v_n)$ in V the following are equivalent:*

(i) *$\mathcal{B}$ is a basis of V;*

(ii) *$\mathcal{B}$ is a* minimal *generating family: it generates V, but no proper subfamily does;*

(iii) *$\mathcal{B}$ is a* maximal *linearly independent family: it is linearly independent, but $(v_1, \dots, v_n, v)$ is dependent for every $v \in V$;*

(iv) *every $v \in V$ can be written in a* unique *way as $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$.*

Proof. (i)⇒(iv): Existence holds as $\mathcal{B}$ generates V. If $\sum \lambda_i v_i = \sum \mu_i v_i$, then $\sum(\lambda_i - \mu_i)v_i = 0$, so $\lambda_i = \mu_i$ for all i by independence.

(iv)⇒(i): $\mathcal{B}$ generates, and $0 = \sum 0 \cdot v_i$ is the unique representation of 0.

(i)⇒(ii): If, say, $(v_1, \dots, v_{n-1})$ generated V, then v_n would be a linear combination of $v_1, \dots, v_{n-1}$, contradicting independence by Lemma 4.3.3.

(ii)⇒(i): If $\mathcal{B}$ were dependent, some v_k would lie in the span of the others by Lemma 4.3.3; then the others would already generate V, contradicting minimality.

(i)⇒(iii): Every v is a linear combination of $\mathcal{B}$, so $(v_1, \dots, v_n, v)$ is dependent.

(iii)⇒(i): For $v \in V$, the family $(v_1, \dots, v_n, v)$ is dependent while $(v_1, \dots, v_n)$ is not, so $v \in \operatorname{span}(\mathcal{B})$ by Lemma 4.3.3. ∎

The scalars $\lambda_1, \dots, \lambda_n$ in (iv) are the *coordinates* of v with respect to $\mathcal{B}$. Note that a basis is an *ordered* family: the coordinate vector $(\lambda_1, \dots, \lambda_n)^\mathsf{T} \in K^n$ depends on the order.

Corollary 4.3.8. *Every finitely generated vector space has a finite basis. More precisely, every finite generating family contains a basis.*

Proof. Remove vectors from a finite generating family as long as the remaining family still generates V. This terminates after finitely many steps with a minimal generating family, which is a basis by Theorem 4.3.7. ∎

Remark 4.3.9 (*). Using Zorn's lemma one shows that *every* vector space has a basis. For instance, $\mathbb{R}$ has a basis over $\mathbb{Q}$ (a so-called Hamel basis), but nobody can write one down explicitly.

4.4 Dimension

The following result is the key to the notion of dimension.

Theorem 4.4.1 (Steinitz exchange lemma)**.** *Let $(v_1, \dots, v_n)$ generate V and let $(w_1, \dots, w_m)$ be linearly independent in V. Then $m \le n$, and after renumbering the v_i, the family $(w_1, \dots, w_m, v_{m+1}, \dots, v_n)$ generates V.*

Proof. Induction on m; the case $m = 0$ is trivial. Let $m \geq 1$. By induction hypothesis applied to $(w_1, \dots, w_{m-1})$ we have $m-1 \leq n$ and, after renumbering, $(w_1, \dots, w_{m-1}, v_m, \dots, v_n)$ generates V. Write

$$w_m = \mu_1 w_1 + \cdots + \mu_{m-1} w_{m-1} + \lambda_m v_m + \cdots + \lambda_n v_n.$$

If $m - 1 = n$ or $\lambda_m = \cdots = \lambda_n = 0$, then $w_m \in \operatorname{span}(w_1, \dots, w_{m-1})$, contradicting the independence of $(w_1, \dots, w_m)$. Hence $m \leq n$, and after renumbering $v_m, \dots, v_n$ we may assume $\lambda_m \neq 0$. Then

$$v_m = \lambda_m^{-1}\big(w_m - \mu_1 w_1 - \cdots - \mu_{m-1} w_{m-1} - \lambda_{m+1} v_{m+1} - \cdots - \lambda_n v_n\big)$$

lies in $\operatorname{span}(w_1, \dots, w_m, v_{m+1}, \dots, v_n)$. This span therefore contains all of $w_1, \dots, w_{m-1}, v_m, \dots, v_n$, and thus equals V. ∎

Corollary 4.4.2. *If V has a finite basis, then all bases of V are finite and have the same number of elements.*

Proof. Let $(v_1, \dots, v_n)$ be a basis. By Theorem 4.4.1, any linearly independent family has at most n elements; in particular every basis $(w_1, \dots, w_m)$ is finite with $m \leq n$. By symmetry, $n \leq m$. ∎

Definition 4.4.3. If V has a finite basis, its number of elements is the *dimension* $\dim V = \dim_K V$, and V is *finite-dimensional*. Otherwise we write $\dim V = \infty$.

Examples 4.4.4. $\dim K^n = n$, $\dim K^{m\times n} = mn$, $\dim K[x]_{\leq d} = d+1$, $\dim\{0\} = 0$, $\dim_{\mathbb{R}} \mathbb{C} = 2$ but $\dim_{\mathbb{C}} \mathbb{C} = 1$, and $\dim K[x] = \infty$. Also $\dim_{\mathbb{Q}} \mathbb{R} = \infty$, since $\mathbb{Q}$ is countable but $\mathbb{R}$ is not.

Theorem 4.4.5 (Basis extension theorem)**.** *Let $\dim V = n < \infty$.*

(a) Every linearly independent family in V has at most n elements and can be extended to a basis of V.

(b) Every generating family has at least n elements, and a family of exactly n vectors is a basis if it is linearly independent or *if it generates V.*

Proof. (a) The bound holds by Theorem 4.4.1. Given an independent family $(w_1, \dots, w_m)$, apply Theorem 4.4.1 with a basis $(v_1, \dots, v_n)$: the family $(w_1, \dots, w_m, v_{m+1}, \dots, v_n)$ generates V and has n elements. By Corollary 4.3.8 it contains a basis, which must have n elements, so it is itself a basis. (b) A generating family contains a basis, so it has at least n elements, and if it has exactly n, it is a basis. An independent family with n elements extends to a basis with n elements by (a), so no vector is added. ∎

Theorem 4.4.6. *Let U be a subspace of a finite-dimensional space V. Then U is finite-dimensional, $\dim U \leq \dim V$, and $\dim U = \dim V$ holds if and only if $U = V$.*

Proof. By Theorem 4.4.1, independent families in U have at most $\dim V$ elements. Choose an independent family $(u_1, \dots, u_m)$ in U with m maximal; by Theorem 4.3.7(iii) it is a basis of U, so $\dim U = m \leq \dim V$. If $m = \dim V$, then $(u_1, \dots, u_m)$ is a basis of V by Theorem 4.4.5(b), hence $U = V$. ∎

Caution 4.4.7. In infinite dimension the last statement fails: $\{p \in K[x] : p(0) = 0\} = \operatorname{span}(x, x^2, \dots)$ is a proper subspace of $K[x]$, but both are infinite-dimensional. Similarly, $V \cong U$ does not imply $U = V$ (shift maps).

4.5 Sums and direct sums

Definition 4.5.1. For subspaces $U_1, \dots, U_k$ of V, the *sum* is $U_1 + \cdots + U_k := \{\, u_1 + \cdots + u_k \mid u_i \in U_i \,\} = \operatorname{span}(U_1 \cup \cdots \cup U_k)$. The sum is *direct*, written $U_1 \oplus \cdots \oplus U_k$, if every element of it has a *unique* representation $u_1 + \cdots + u_k$ with $u_i \in U_i$.

Lemma 4.5.2. *The sum $U_1 + \cdots + U_k$ is direct if and only if $u_1 + \cdots + u_k = 0$ with $u_i \in U_i$ implies $u_1 = \cdots = u_k = 0$. For $k = 2$ this is equivalent to $U_1 \cap U_2 = \{0\}$.*

Proof. The first statement follows as in the proof of Theorem 4.3.7 (uniqueness of the representation of 0 suffices, by taking differences). For $k = 2$: if $0 \neq u \in U_1 \cap U_2$, then $u + (-u) = 0$ is a non-trivial representation. Conversely, if $u_1 + u_2 = 0$, then $u_1 = -u_2 \in U_1 \cap U_2 = \{0\}$. ∎

Caution 4.5.3. For $k \geq 3$ it is *not* sufficient that $U_i \cap U_j = \{0\}$ for all $i \neq j$: three distinct lines through 0 in $\mathbb{R}^2$ intersect pairwise trivially, but their sum is not direct.

Theorem 4.5.4 (Dimension formula for sums)**.** *Let U, W be finite-dimensional subspaces of V. Then*

$$\dim(U + W) + \dim(U \cap W) = \dim U + \dim W.$$

Proof. Let $(v_1, \dots, v_r)$ be a basis of $U \cap W$. Extend it to a basis $(v_1, \dots, v_r, u_1, \dots, u_s)$ of U and to a basis $(v_1, \dots, v_r, w_1, \dots, w_t)$ of W (Theorem 4.4.5). We claim that $\mathcal{B} = (v_1, \dots, v_r, u_1, \dots, u_s, w_1, \dots, w_t)$ is a basis of $U + W$; then $\dim(U + W) = r + s + t = (r + s) + (r + t) - r$, as claimed.

Clearly $\mathcal{B}$ generates $U + W$. Suppose $\sum_i \alpha_i v_i + \sum_j \beta_j u_j + \sum_k \gamma_k w_k = 0$. Then

$$w := \sum_k \gamma_k w_k = -\sum_i \alpha_i v_i - \sum_j \beta_j u_j \in U \cap W,$$

so $w = \sum_i \delta_i v_i$ for some $\delta_i \in K$. Then $\sum_k \gamma_k w_k - \sum_i \delta_i v_i = 0$, and independence of the basis of W yields $\gamma_k = 0$ for all k. Now $\sum_i \alpha_i v_i + \sum_j \beta_j u_j = 0$ implies $\alpha_i = \beta_j = 0$ by independence of the basis of U. ∎

Example 4.5.5. Two distinct planes $U \neq W$ through the origin in $\mathbb{R}^3$ satisfy $U + W = \mathbb{R}^3$ (as $U \subsetneq U + W$, so $\dim(U + W) = 3$), hence $\dim(U \cap W) = 2 + 2 - 3 = 1$: they intersect in a line.

Corollary 4.5.6. *If V is finite-dimensional and $U \subseteq V$ is a subspace, then there exists a* complement *W, i.e. a subspace with $V = U \oplus W$. For any such W, $\dim W = \dim V - \dim U$.*

Proof. Extend a basis $(u_1, \dots, u_m)$ of U to a basis $(u_1, \dots, u_m, w_1, \dots, w_k)$ of V and put $W := \operatorname{span}(w_1, \dots, w_k)$. Then $U + W = V$ and $U \cap W = \{0\}$ by uniqueness of coordinates. The dimension statement follows from Theorem 4.5.4. ∎

Complements are not unique: every line in $\mathbb{R}^2$ other than U itself is a complement of a given line U through 0.

4.6 Computing with subspaces of K^n

In practice, subspaces of K^n are given in one of two ways: *explicitly*, as the span of finitely many vectors, or *implicitly*, as the solution set of a homogeneous system $Ax = 0$. Gaussian elimination converts between the two descriptions and solves the standard tasks.

Task 1: a basis of $U = \operatorname{span}(u_1, \dots, u_k)$. Write $u_1^{\mathsf{T}}, \dots, u_k^{\mathsf{T}}$ as the *rows* of a matrix and bring it to row echelon form. Row operations do not change the span of the rows (each new row is a linear combination of the old ones and vice versa), and the non-zero rows of an echelon form are linearly independent. Hence they form a basis of U, and $\dim U$ is the number of non-zero rows.

Task 2: a basis of $L(A, 0)$. Bring A to reduced row echelon form; for each free variable x_k the solution with $x_k = 1$ and all other free variables 0 yields a basis vector (cf. Example 3.3.2). There are exactly $n - r$ of them, and they are independent since each has a 1 in a position where all others have a 0.

Task 3: from explicit to implicit. Given $U = \operatorname{span}(u_1, \dots, u_k)$, a vector y lies in U iff the system $\sum_j \lambda_j u_j = y$ in the unknowns λ_j is solvable. Eliminating with a *symbolic* right-hand side y produces the solvability conditions $\tilde{b}_{r+1}(y) = \cdots = \tilde{b}_m(y) = 0$ of Theorem 3.3.1, which are linear equations in y describing U.

Task 4: intersections and sums. $U + W$ is spanned by the union of spanning sets (Task 1). For $U \cap W$, describe both implicitly (Task 3) and solve the combined system (Task 2); alternatively, solve $\sum \lambda_i u_i = \sum \mu_j w_j$ for (λ, μ).

Example 4.6.1. In $\mathbb{R}^4$ let $U = \operatorname{span}(u_1, u_2, u_3)$ with $u_1 = (1, 1, 0, 2)^{\mathsf{T}}$, $u_2 = (2, 1, 1, 3)^{\mathsf{T}}$, $u_3 = (0, 1, -1, 1)^{\mathsf{T}}$, and let $W = \{\, x \in \mathbb{R}^4 \mid x_1 - x_2 + x_4 = 0 \,\}$.

Basis of U. Row reduction

$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 2 & 1 & 1 & 3 \\ 0 & 1 & -1 & 1 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

shows $\dim U = 2$ with basis $\big((1, 1, 0, 2)^{\mathsf{T}}, (0, -1, 1, -1)^{\mathsf{T}}\big)$. (Indeed $u_3 = 2u_1 - u_2$.)

Implicit description of U. $y = \alpha(1, 1, 0, 2)^{\mathsf{T}} + \beta(0, -1, 1, -1)^{\mathsf{T}}$ means $y_1 = \alpha$, $y_3 = \beta$, $y_2 = \alpha - \beta$, $y_4 = 2\alpha - \beta$. Eliminating α, β gives the two equations

$$y_2 = y_1 - y_3, \qquad y_4 = 2y_1 - y_3.$$

Hence $U = \{\, y \mid y_1 - y_2 - y_3 = 0,\ 2y_1 - y_3 - y_4 = 0 \,\}$, as expected from $\dim U = 4 - 2$.

Intersection. $U \cap W$ is the solution set of the three equations $y_1 - y_2 - y_3 = 0$, $2y_1 - y_3 - y_4 = 0$, $y_1 - y_2 + y_4 = 0$. Subtracting the first from the third gives $y_3 + y_4 = 0$, then the second gives $2y_1 = y_3 + y_4 = 0$. So $y_1 = 0$, $y_4 = -y_3$, $y_2 = -y_3$, and $U \cap W = \operatorname{span}\big((0, -1, 1, -1)^{\mathsf{T}}\big)$. By Theorem 4.5.4, $\dim(U + W) = 2 + 3 - 1 = 4$, i.e. $U + W = \mathbb{R}^4$.

Chapter 5

Linear Maps and Matrices

5.1 Linear maps

Definition 5.1.1. Let V, W be K-vector spaces. A map $f\colon V \to W$ is *linear* (a *homomorphism* of vector spaces) if for all $v, v' \in V$ and $\lambda \in K$

$$f(v+v') = f(v) + f(v') \qquad \text{and} \qquad f(\lambda v) = \lambda f(v).$$

A bijective linear map is an *isomorphism*; if one exists, V and W are *isomorphic*, $V \cong W$. A linear map $V \to V$ is an *endomorphism*, a bijective one an *automorphism*. We write $\mathrm{Hom}_K(V,W)$ for the set of linear maps $V \to W$ and $\mathrm{End}_K(V) = \mathrm{Hom}_K(V,V)$.

Linear maps are in particular homomorphisms of the groups $(V,+) \to (W,+)$, so $f(0) = 0$ and $f(-v) = -f(v)$ by Lemma 2.1.5. By induction, $f(\sum \lambda_i v_i) = \sum \lambda_i f(v_i)$.

Examples 5.1.2. (1) For $A \in K^{m\times n}$ the map $f_A\colon K^n \to K^m$, $x \mapsto Ax$, is linear (Lemma 3.1.2).

(2) Rotation of $\mathbb{R}^2$ about 0 by the angle φ is linear; it is f_A with $A = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}$.

(3) Differentiation $D\colon K[x] \to K[x]$, $\sum a_k x^k \mapsto \sum k a_k x^{k-1}$, is linear. So is $C^1(\mathbb{R}) \to C(\mathbb{R})$, $f \mapsto f'$, and $C([0,1]) \to \mathbb{R}$, $f \mapsto \int_0^1 f(t)\,dt$.

(4) The *evaluation* $K[x] \to K$, $p \mapsto p(a)$, is linear for every $a \in K$.

(5) The *transposition* $K^{m\times n} \to K^{n\times m}$, $A \mapsto A^{\mathsf{T}}$, where $(A^{\mathsf{T}})_{ij} = a_{ji}$, is an isomorphism.

(6) $x \mapsto x+1$ on $\mathbb{R}$ is not linear ($0 \mapsto 1$), and neither is $x \mapsto x^2$.

Lemma 5.1.3. *Let $f\colon V \to W$ and $g\colon W \to U$ be linear.*

(a) $g \circ f$ is linear. If f is an isomorphism, so is f^{-1}.

(b) $\mathrm{Hom}_K(V,W)$ is a subspace of W^V; in particular, sums and scalar multiples of linear maps are linear.

Proof. (a) Linearity of $g \circ f$ is clear. For $w, w' \in W$ put $v = f^{-1}(w)$, $v' = f^{-1}(w')$; then $f(v+v') = w + w'$, so $f^{-1}(w+w') = v + v' = f^{-1}(w) + f^{-1}(w')$; likewise for scalars. (b) Direct verification. ∎

Definition 5.1.4. For linear $f\colon V \to W$, the *kernel* $\ker f := \{\, v \in V \mid f(v) = 0 \,\}$ and the *image* $\mathrm{im}\, f := f(V)$ are subspaces of V and W, respectively. The *rank* of f is $\mathrm{rank}\, f := \dim \mathrm{im}\, f$.

By Lemma 2.1.5, *f is injective if and only if* $\ker f = \{0\}$. For $f = f_A$, the kernel is the solution set $L(A,0)$, and the image is spanned by the columns of A, because $Ax = \sum_j x_j a_j$ where a_j denotes the j-th column.

Theorem 5.1.5 (Linear maps are determined by a basis)**.** *Let $(v_1, \ldots, v_n)$ be a basis of V and $w_1, \ldots, w_n \in W$ arbitrary. Then there is exactly one linear map $f\colon V \to W$ with $f(v_i) = w_i$ for $i = 1, \ldots, n$. Moreover:*

(a) $\mathrm{im}\, f = \mathrm{span}(w_1, \ldots, w_n)$;

(b) f is injective $\iff$ $(w_1, \ldots, w_n)$ is linearly independent;

(c) f is an isomorphism $\iff$ $(w_1, \ldots, w_n)$ is a basis of W.

Proof. Uniqueness: a linear f must satisfy $f(\sum \lambda_i v_i) = \sum \lambda_i w_i$. Existence: since coordinates are unique (Theorem 4.3.7), this formula defines a map $V \to W$, which is easily seen to be linear. (a) is clear. (b) $\sum \lambda_i w_i = 0 \iff f(\sum \lambda_i v_i) = 0$; so $\ker f = \{0\}$ iff only the trivial combination of the w_i vanishes. (c) follows from (a) and (b). ∎

Corollary 5.1.6. *(a) If $\mathcal{B} = (v_1, \dots, v_n)$ is a basis of V, then $\Phi_{\mathcal{B}} \colon K^n \to V$, $(\lambda_1, \dots, \lambda_n)^{\mathsf{T}} \mapsto \sum \lambda_i v_i$, is an isomorphism (the* coordinate isomorphism*).*

(b) Finite-dimensional vector spaces V, W are isomorphic if and only if $\dim V = \dim W$.

Thus every n-dimensional K-vector space "is" K^n – but not canonically: the identification depends on the choice of a basis. Much of linear algebra consists in choosing bases cleverly.

5.2 The rank–nullity theorem

Theorem 5.2.1 (Rank–nullity theorem)**.** *Let V be finite-dimensional and $f \colon V \to W$ linear. Then*

$$\dim V = \dim \ker f + \dim \operatorname{im} f.$$

Proof. Choose a basis $(u_1, \dots, u_k)$ of $\ker f$ and extend it to a basis $(u_1, \dots, u_k, v_1, \dots, v_r)$ of V. We show that $(f(v_1), \dots, f(v_r))$ is a basis of $\operatorname{im} f$; then $\dim V = k + r$ is the claim.

Generating: for $v = \sum \alpha_i u_i + \sum \beta_j v_j$ we get $f(v) = \sum \beta_j f(v_j)$, since $f(u_i) = 0$. *Independent:* if $\sum \beta_j f(v_j) = 0$, then $\sum \beta_j v_j \in \ker f$, so $\sum \beta_j v_j = \sum \alpha_i u_i$ for some α_i. Independence of the basis of V forces all β_j (and α_i) to vanish. ∎

Corollary 5.2.2. *Let $\dim V = \dim W < \infty$ and $f \colon V \to W$ linear. Then*

$$f \text{ injective} \iff f \text{ surjective} \iff f \text{ bijective.}$$

Proof. f injective $\iff \dim \ker f = 0 \iff \dim \operatorname{im} f = \dim W \iff \operatorname{im} f = W$, using Theorem 5.2.1 and Theorem 4.4.6. ∎

Caution 5.2.3. This fails in infinite dimensions: on $K[x]$, the map $p \mapsto xp$ is injective but not surjective, and differentiation is surjective (if $\operatorname{char} K = 0$) but not injective.

Corollary 5.2.4. *Let $A \in K^{m \times n}$ and $r = \dim \operatorname{im} f_A$. Then $\dim L(A, 0) = n - r$. In particular, the number of free variables in Gaussian elimination equals $n - r$ and does not depend on how the elimination was performed.*

5.3 The matrix of a linear map

Let V, W be finite-dimensional with bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$. For $f \in \operatorname{Hom}_K(V, W)$ there are unique $a_{ij} \in K$ with

$$f(v_j) = \sum_{i=1}^{m} a_{ij} w_i \qquad (j = 1, \dots, n). \tag{5.1}$$

Definition 5.3.1. The matrix $M_{\mathcal{C}}^{\mathcal{B}}(f) := (a_{ij}) \in K^{m \times n}$ is the *matrix of f with respect to $\mathcal{B}$ and $\mathcal{C}$*. Mnemonic: *the j-th column contains the coordinates of the image of the j-th basis vector.*

Theorem 5.3.2. *The map $\operatorname{Hom}_K(V, W) \to K^{m \times n}$, $f \mapsto M_{\mathcal{C}}^{\mathcal{B}}(f)$, is an isomorphism of vector spaces. In particular $\dim \operatorname{Hom}_K(V, W) = \dim V \cdot \dim W$.*

Proof. Linearity is immediate from (5.1). Bijectivity is a reformulation of Theorem 5.1.5: the images $f(v_j)$, i.e. the columns, can be prescribed arbitrarily and determine f. ∎

For the standard bases $\mathcal{E}$ of K^n and K^m we get $M_{\mathcal{E}}^{\mathcal{E}}(f_A) = A$, since $f_A(e_j) = Ae_j$ is the j-th column of A.

Definition 5.3.3. For $A \in K^{l\times m}$ and $B \in K^{m\times n}$ the *product* $AB \in K^{l\times n}$ is defined by

$$(AB)_{ik} := \sum_{j=1}^{m} a_{ij}b_{jk} \qquad (1 \le i \le l,\ 1 \le k \le n).$$

Theorem 5.3.4. *Let $f\colon V \to W$, $g\colon W \to U$ be linear and $\mathcal{B},\mathcal{C},\mathcal{D}$ bases of V, W, U. Then*

$$M_{\mathcal{D}}^{\mathcal{B}}(g \circ f) = M_{\mathcal{D}}^{\mathcal{C}}(g) \cdot M_{\mathcal{C}}^{\mathcal{B}}(f).$$

Proof. Let $A = M_{\mathcal{C}}^{\mathcal{B}}(f)$, $B = M_{\mathcal{D}}^{\mathcal{C}}(g)$, and write $\mathcal{D} = (u_1, \dots, u_l)$. Then

$$g(f(v_k)) = g\Big(\sum_j a_{jk}w_j\Big) = \sum_j a_{jk}\sum_i b_{ij}u_i = \sum_i \Big(\sum_j b_{ij}a_{jk}\Big)u_i = \sum_i (BA)_{ik}u_i. \qquad ∎$$

Corollary 5.3.5. *Matrix multiplication is associative and distributive: $(AB)C = A(BC)$, $A(B+B') = AB + AB'$, etc., whenever the products are defined. $K^{n\times n}$ is a ring with unit $I_n = (\delta_{ij})$, which is non-commutative for $n \ge 2$.*

Proof. Associativity of composition of maps translates into associativity of matrix multiplication via Theorem 5.3.4 (applied to f_A, f_B, f_C). ∎

Moreover, if $x \in K^n$ is the coordinate vector of $v \in V$ with respect to $\mathcal{B}$, then $M_{\mathcal{C}}^{\mathcal{B}}(f)\,x$ is the coordinate vector of $f(v)$ with respect to $\mathcal{C}$. In other words, the diagram

$$\begin{array}{ccc} K^n & \xrightarrow{x\mapsto Ax} & K^m \\ {\scriptstyle\Phi_{\mathcal{B}}}\downarrow\cong & & \cong\downarrow{\scriptstyle\Phi_{\mathcal{C}}} \\ V & \xrightarrow{\ f\ } & W \end{array} \qquad A = M_{\mathcal{C}}^{\mathcal{B}}(f),$$

commutes, i.e. $f \circ \Phi_{\mathcal{B}} = \Phi_{\mathcal{C}} \circ f_A$.

Definition 5.3.6. $A \in K^{n\times n}$ is *invertible* if there is $B \in K^{n\times n}$ with $AB = BA = I_n$; then $B = A^{-1}$ is unique. The invertible $n \times n$ matrices form a group $\mathrm{GL}_n(K)$, the *general linear group*.

Proposition 5.3.7. *For $A \in K^{n\times n}$ the following are equivalent: (i) A is invertible; (ii) f_A is an isomorphism; (iii) $Ax = 0$ has only the solution $x = 0$; (iv) the columns of A form a basis of K^n; (v) there is B with $BA = I_n$; (vi) there is B with $AB = I_n$.*

Proof. (i)⇔(ii) by Theorems 5.3.2 and 5.3.4; (ii)⇔(iii)⇔(iv) by Corollary 5.2.2 and Theorem 5.1.5. If $BA = I_n$, then $f_B \circ f_A = \mathrm{id}$, so f_A is injective, hence bijective by Corollary 5.2.2. If $AB = I_n$, then f_A is surjective, hence bijective. ∎

Example 5.3.8 (Computing the inverse)**.** To invert A, apply Gaussian elimination to $(A \mid I_n)$. If A is invertible, the reduced row echelon form is $(I_n \mid A^{-1})$; otherwise a zero row appears on the left. (Reason: each row operation is left multiplication by an invertible *elementary matrix*; if $E_k \cdots E_1 A = I_n$, then $E_k \cdots E_1 = A^{-1}$.) For instance, over $\mathbb{R}$:

$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array}\right) \rightsquigarrow \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{5}{2} & 1 \end{array}\right) \rightsquigarrow \left(\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -5 & 2 \end{array}\right),$$

so $\left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right)$.

5.4 Change of basis

Let $\mathcal{B}, \mathcal{B}'$ be two bases of V. The *transition matrix* from $\mathcal{B}'$ to $\mathcal{B}$ is

$$T_{\mathcal{B}}^{\mathcal{B}'} := M_{\mathcal{B}}^{\mathcal{B}'}(\mathrm{id}_V) \in \mathrm{GL}_n(K);$$

its j-th column contains the $\mathcal{B}$-coordinates of the j-th vector of $\mathcal{B}'$. If x' are the $\mathcal{B}'$-coordinates of v, then $T_{\mathcal{B}}^{\mathcal{B}'} x'$ are its $\mathcal{B}$-coordinates. By Theorem 5.3.4, $T_{\mathcal{B}'}^{\mathcal{B}} = (T_{\mathcal{B}}^{\mathcal{B}'})^{-1}$.

Theorem 5.4.1 (Transformation formula)**.** *Let $f\colon V \to W$ be linear, $\mathcal{B}, \mathcal{B}'$ bases of V and $\mathcal{C}, \mathcal{C}'$ bases of W. Then*

$$M_{\mathcal{C}'}^{\mathcal{B}'}(f) = T_{\mathcal{C}'}^{\mathcal{C}} \cdot M_{\mathcal{C}}^{\mathcal{B}}(f) \cdot T_{\mathcal{B}}^{\mathcal{B}'}.$$

In particular, for an endomorphism f of V and $S := T_{\mathcal{B}}^{\mathcal{B}'}$: $M_{\mathcal{B}'}^{\mathcal{B}'}(f) = S^{-1} M_{\mathcal{B}}^{\mathcal{B}}(f)\, S$.

Proof. Apply Theorem 5.3.4 to $f = \mathrm{id}_W \circ f \circ \mathrm{id}_V$. ∎

Definition 5.4.2. Matrices $A, B \in K^{n\times n}$ are *similar* if $B = S^{-1}AS$ for some $S \in \mathrm{GL}_n(K)$; $A, B \in K^{m\times n}$ are *equivalent* if $B = TAS$ for some $T \in \mathrm{GL}_m(K)$, $S \in \mathrm{GL}_n(K)$.

So two matrices are similar iff they describe the same endomorphism with respect to two (possibly different) bases. Classifying matrices up to similarity is a deep problem (Jordan normal form, Linear Algebra II); up to equivalence it is easy:

Theorem 5.4.3 (Normal form under equivalence)**.** *Let $f\colon V \to W$ be linear of rank r, $\dim V = n$, $\dim W = m$. There are bases $\mathcal{B}$ of V and $\mathcal{C}$ of W with*

$$M_{\mathcal{C}}^{\mathcal{B}}(f) = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \in K^{m\times n}.$$

Consequently, two $m \times n$ matrices are equivalent iff they have the same rank.

Proof. As in the proof of Theorem 5.2.1, choose a basis $\mathcal{B} = (v_1, \dots, v_r, u_1, \dots, u_{n-r})$ of V with $(u_1, \dots, u_{n-r})$ a basis of $\ker f$. Then $w_i := f(v_i)$, $i = 1, \dots, r$, form a basis of $\operatorname{im} f$, which we extend to a basis $\mathcal{C}$ of W. ∎

5.5 The rank of a matrix

Definition 5.5.1. The *column rank* of $A \in K^{m\times n}$ is the dimension of the span of its columns in K^m, i.e. $\operatorname{rank} f_A$; the *row rank* is the dimension of the span of its rows in $K^{1\times n}$.

Theorem 5.5.2. *Row rank and column rank of a matrix coincide. This number is called the* rank $\operatorname{rank} A$*, and* $\operatorname{rank} A = \operatorname{rank} A^{\mathsf{T}}$.

Proof. Multiplying A by invertible matrices from the left or right does not change the column rank (the image of f_{TAS} is T applied to the image of f_A, and f_T is an isomorphism), nor the row rank (apply the same argument to A^{T}, using $(TAS)^{\mathsf{T}} = S^{\mathsf{T}}A^{\mathsf{T}}T^{\mathsf{T}}$ and that S^{T} is invertible with inverse $(S^{-1})^{\mathsf{T}}$). By Theorem 5.4.3, $A = T \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) S$ with invertible T, S, and for the normal form both ranks equal r. ∎

Remark 5.5.3. In practice, $\operatorname{rank} A$ is the number r of non-zero rows of a row echelon form of A: row operations do not change the row space, and the non-zero rows of an echelon form are clearly linearly independent.

Corollary 5.5.4 (Solvability criterion)**.** *The system $Ax = b$ is solvable if and only if* $\operatorname{rank} A = \operatorname{rank}(A \mid b)$*. It is uniquely solvable for every b iff A is square and invertible.*

5.6 Quotient spaces*

Let $U \subseteq V$ be a subspace. The relation $v \sim v' \iff v - v' \in U$ is an equivalence relation on V; the class of v is the *affine subspace* $v + U$. On $V/U := V/\!\sim$ we define $(v+U)+(v'+U) := (v+v')+U$ and $\lambda(v+U) := \lambda v + U$; exactly as for $\mathbb{Z}/n\mathbb{Z}$ one checks that this is well defined, and V/U becomes a vector space, the *quotient space*. The canonical projection $\pi\colon V \to V/U$, $v \mapsto v+U$ is linear and surjective with $\ker \pi = U$. By Theorem 5.2.1, $\dim V/U = \dim V - \dim U$ if V is finite-dimensional.

Theorem 5.6.1 (Homomorphism theorem)**.** *For linear $f\colon V \to W$, the map $\bar{f}\colon V/\ker f \to \operatorname{im} f$, $v + \ker f \mapsto f(v)$, is a well-defined isomorphism.*

Proof. $v + \ker f = v' + \ker f$ implies $f(v) - f(v') = f(v - v') = 0$, so $\bar{f}$ is well defined; it is clearly linear and surjective, and $\bar{f}(v + \ker f) = 0$ means $v \in \ker f$, i.e. $v + \ker f$ is the zero of $V/\ker f$. ∎

5.7 Elementary matrices

Row operations can be expressed by matrix multiplication, which makes many arguments about Gaussian elimination rigorous. For $i \neq k$ and $\lambda \in K$ let

$$P_{ik} := I_m - E_{ii} - E_{kk} + E_{ik} + E_{ki}, \qquad D_i(\lambda) := I_m + (\lambda - 1)E_{ii}, \qquad T_{ik}(\lambda) := I_m + \lambda E_{ik},$$

where E_{ik} are the matrix units. These are the *elementary matrices*. A direct computation shows that for $A \in K^{m\times n}$

- $P_{ik}A$ arises from A by interchanging rows i and k (R1)
- $D_i(\lambda)A$ arises by multiplying row i by λ (R2)
- $T_{ik}(\lambda)A$ arises by adding λ times row k to row i, (R3)

and that multiplication *from the right* performs the corresponding *column* operations. Elementary matrices are invertible: $P_{ik}^{-1} = P_{ik}$, $D_i(\lambda)^{-1} = D_i(\lambda^{-1})$ for $\lambda \neq 0$, and $T_{ik}(\lambda)^{-1} = T_{ik}(-\lambda)$. Their determinants (Chapter 6) are -1, λ and 1, respectively.

Theorem 5.7.1. *Every invertible matrix is a product of elementary matrices. More precisely, for every $A \in K^{m\times n}$ there is a product $E = E_k \cdots E_1$ of elementary matrices such that EA is in reduced row echelon form, and $A \in \mathrm{GL}_n(K)$ if and only if $EA = I_n$.*

Proof. The first statement is Theorem 3.2.4 rephrased. If A is invertible, so is EA; an invertible matrix in reduced row echelon form has no zero row, hence a pivot in every column, and is therefore I_n. Then $A = E^{-1} = E_1^{-1}\cdots E_k^{-1}$ is a product of elementary matrices. Conversely, $EA = I_n$ implies $A = E^{-1}$. ∎

Remark 5.7.2 (LR decomposition). If Gaussian elimination on $A \in \mathrm{GL}_n(K)$ succeeds without row interchanges, only matrices $T_{ik}(\lambda)$ with $i > k$ are used; their product is lower triangular with ones on the diagonal. One obtains $A = LR$ with L lower triangular (unipotent) and R upper triangular. Once L and R are known, $Ax = b$ is solved for many right-hand sides by forward and backward substitution with only $O(n^2)$ operations each. In general one needs a permutation: $PA = LR$.

5.8 The dual space

Definition 5.8.1. The *dual space* of a K-vector space V is $V^* := \mathrm{Hom}_K(V, K)$. Its elements are called *linear forms* (or linear functionals). For a linear map $f\colon V \to W$, the *dual map* is $f^*\colon W^* \to V^*$, $\varphi \mapsto \varphi \circ f$.

Examples of linear forms are the evaluation $p \mapsto p(a)$ on $K[x]$, the trace on $K^{n\times n}$, integration $f \mapsto \int_0^1 f$ on $C([0,1])$, and on K^n the maps $x \mapsto a_1x_1 + \cdots + a_nx_n$ given by a row vector $(a_1, \ldots, a_n)$. In fact the last example is the general one for $V = K^n$, by Theorem 5.3.2.

Proposition 5.8.2 (Dual basis)**.** *Let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis of V. For $i = 1, \ldots, n$ let $v_i^* \in V^*$ be the linear form with $v_i^*(v_j) = \delta_{ij}$ (Theorem 5.1.5). Then $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ is a basis of V^*, the* dual basis*; in particular* $\dim V^* = \dim V$*. Moreover $v_i^*(v)$ is the i-th coordinate of v with respect to $\mathcal{B}$, and every $\varphi \in V^*$ satisfies $\varphi = \sum_i \varphi(v_i)\, v_i^*$.*

Proof. For $v = \sum_j \lambda_j v_j$ we have $v_i^*(v) = \sum_j \lambda_j \delta_{ij} = \lambda_i$. The linear forms φ and $\sum_i \varphi(v_i) v_i^*$ agree on the basis, hence are equal; so $\mathcal{B}^*$ generates V^*. If $\sum_i \mu_i v_i^* = 0$, evaluating at v_j gives $\mu_j = 0$. ∎

Proposition 5.8.3. *Let $f\colon V \to W$ be linear, $\mathcal{B}, \mathcal{C}$ bases of V, W (finite-dimensional). Then*

$$M_{\mathcal{B}^*}^{\mathcal{C}^*}(f^*) = \big(M_{\mathcal{C}}^{\mathcal{B}}(f)\big)^{\mathsf{T}}.$$

Proof. Let $A = M_{\mathcal{C}}^{\mathcal{B}}(f)$, $\mathcal{B} = (v_j)$, $\mathcal{C} = (w_i)$. By Proposition 5.8.2, the j-th coordinate of $f^*(w_i^*)$ with respect to $\mathcal{B}^*$ is $f^*(w_i^*)(v_j) = w_i^*(f(v_j)) = w_i^*\big(\sum_k a_{kj} w_k\big) = a_{ij}$. So the i-th column of the matrix of f^* is the i-th row of A. ∎

This gives a conceptual explanation of the transpose and, combined with the rank–nullity theorem, a second proof of Theorem 5.5.2 (see the exercises of Linear Algebra II, where also the *bidual* V^{**} and annihilators are studied).

Caution 5.8.4. V and V^* are isomorphic if $\dim V < \infty$, but there is no *canonical* isomorphism: the isomorphism $v_i \mapsto v_i^*$ depends on the choice of the basis. For $V = K[x]$ (countable dimension) one can show that $V^* \cong K^{\mathbb{N}}$ is *not* isomorphic to V.

Chapter 6

Determinants

The determinant assigns to every square matrix A a scalar $\det A$ that is non-zero precisely when A is invertible. Geometrically, for $K = \mathbb{R}$, $|\det A|$ is the volume of the parallelepiped spanned by the columns of A, and the sign of $\det A$ records whether f_A preserves orientation.

6.1 Permutations

Recall that S_n denotes the group of bijections $\sigma\colon [n] \to [n]$. We write $\sigma = \left(\begin{smallmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{smallmatrix}\right)$. A *transposition* $\tau = (i\ j)$, $i \neq j$, interchanges i and j and fixes all other elements; note $\tau^{-1} = \tau$.

Lemma 6.1.1. *Every $\sigma \in S_n$ is a product of transpositions (where the identity is the empty product).*

Proof. Induction on n; $n = 1$ is trivial. If $\sigma(n) = n$, then σ restricts to a permutation of $[n-1]$ and we are done by induction. Otherwise, $\tau = (n\ \sigma(n))$ satisfies $(\tau\sigma)(n) = n$, so $\tau\sigma = \tau_1 \cdots \tau_k$ is a product of transpositions and so is $\sigma = \tau\tau_1 \cdots \tau_k$. ∎

Definition 6.1.2. For $\sigma \in S_n$, a pair (i, j) with $i < j$ and $\sigma(i) > \sigma(j)$ is an *inversion* of σ. The *sign* of σ is $\operatorname{sgn}\sigma := (-1)^{N(\sigma)}$, where $N(\sigma)$ is the number of inversions. σ is *even* if $\operatorname{sgn}\sigma = 1$ and *odd* otherwise.

Lemma 6.1.3. $\operatorname{sgn}\sigma = \displaystyle\prod_{1 \le i < j \le n} \frac{\sigma(j) - \sigma(i)}{j - i}$.

Proof. As $\{i, j\}$ runs through all 2-element subsets of $[n]$, so does $\{\sigma(i), \sigma(j)\}$. Hence numerator and denominator agree up to sign, and a factor -1 arises exactly for each inversion. ∎

Theorem 6.1.4. $\operatorname{sgn}\colon S_n \to \{\pm 1\}$ *is a group homomorphism:* $\operatorname{sgn}(\sigma\rho) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\rho)$. *Every transposition is odd. Consequently, if $\sigma = \tau_1 \cdots \tau_k$ with transpositions τ_i, then $\operatorname{sgn}\sigma = (-1)^k$; in particular the parity of k depends only on σ.*

Proof. By Lemma 6.1.3,

$$\operatorname{sgn}(\sigma\rho) = \prod_{i<j} \frac{\sigma(\rho(j)) - \sigma(\rho(i))}{\rho(j) - \rho(i)} \cdot \prod_{i<j} \frac{\rho(j) - \rho(i)}{j - i}.$$

The second product is $\operatorname{sgn}\rho$. In the first product, each factor is symmetric in i, j, so it equals the product over all 2-element subsets $\{k, l\} = \{\rho(i), \rho(j)\}$, i.e. $\prod_{k<l} \frac{\sigma(l)-\sigma(k)}{l-k} = \operatorname{sgn}\sigma$. A transposition $(i\ j)$ with $i < j$ has exactly $2(j - i) - 1$ inversions, namely (i, j) and $(i, k), (k, j)$ for $i < k < j$. ∎

The even permutations form a subgroup $A_n = \ker \operatorname{sgn}$ (the *alternating group*) with $n!/2$ elements for $n \geq 2$.

6.2 Multilinear alternating forms and the determinant

We regard a matrix $A \in K^{n\times n}$ as the n-tuple of its columns, $A = (a_1|\dots|a_n)$.

Definition 6.2.1. A map $\delta\colon K^{n\times n} \to K$ is a *determinant function* if

(D1) δ is *multilinear*, i.e. linear in each column when the other columns are fixed;

(D2) δ is *alternating*: $\delta(A) = 0$ whenever two columns of A are equal;

(D3) δ is *normalised*: $\delta(I_n) = 1$.

Lemma 6.2.2. *Let δ satisfy (D1) and (D2). Then*

(a) interchanging two columns changes the sign of δ;

(b) adding a multiple of one column to another column does not change δ;

(c) if the columns of A are linearly dependent, then $\delta(A) = 0$;

(d) $\delta(e_{\sigma(1)}|\cdots|e_{\sigma(n)}) = \operatorname{sgn}(\sigma)\,\delta(I_n)$ for all $\sigma \in S_n$.

Proof. (a) Writing only the columns i and j: $0 = \delta(\dots, a+b, \dots, a+b, \dots) = \delta(\dots, a, \dots, b, \dots) + \delta(\dots, b, \dots, a, \dots)$, the two other terms vanishing by (D2). (b) $\delta(\dots, a_i + \lambda a_j, \dots, a_j, \dots) = \delta(A) + \lambda\delta(\dots, a_j, \dots, a_j, \dots) = \delta(A)$. (c) If, say, $a_1 = \sum_{i\geq 2} \lambda_i a_i$, expand by (D1); every summand has two equal columns. (d) Write σ as a product of k transpositions (Lemma 6.1.1), apply (a) k times and use Theorem 6.1.4. ∎

Theorem 6.2.3 (Existence and uniqueness of the determinant)**.** *There is exactly one determinant function* $\det\colon K^{n\times n} \to K$*, given by the* Leibniz formula

$$\det A = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\, a_{\sigma(1)1}\, a_{\sigma(2)2} \cdots a_{\sigma(n)n}.$$

Moreover, every map δ satisfying (D1) and (D2) is of the form $\delta = \delta(I_n)\cdot\det$.

Proof. Uniqueness. Let δ satisfy (D1), (D2). Writing $a_j = \sum_i a_{ij} e_i$ and expanding by multilinearity,

$$\delta(A) = \sum_{i_1,\dots,i_n=1}^{n} a_{i_1 1} \cdots a_{i_n n}\, \delta(e_{i_1}|\cdots|e_{i_n}).$$

By (D2), only terms with pairwise distinct $i_1, \dots, i_n$ survive, i.e. $i_j = \sigma(j)$ for some $\sigma \in S_n$. By Lemma 6.2.2(d), $\delta(A) = \delta(I_n)\sum_\sigma \operatorname{sgn}(\sigma) a_{\sigma(1)1}\cdots a_{\sigma(n)n}$.

Existence. We verify that the Leibniz formula defines a determinant function. (D1): each summand is linear in column j, since it contains exactly one factor $a_{\sigma(j)j}$ from that column. (D3): only $\sigma = \mathrm{id}$ contributes for $A = I_n$. (D2): suppose $a_k = a_l$ with $k < l$, and let $\tau = (k\ l)$. Then $\sigma \mapsto \sigma\tau$ is a bijection from A_n onto the set of odd permutations. Since $a_{ij} = a_{i\tau(j)}$ for all i, j, the summands for σ and $\sigma\tau$ are $\operatorname{sgn}(\sigma)\prod_j a_{\sigma(j)j}$ and

$$-\operatorname{sgn}(\sigma)\prod_j a_{\sigma\tau(j)j} = -\operatorname{sgn}(\sigma)\prod_j a_{\sigma\tau(j)\tau(j)} = -\operatorname{sgn}(\sigma)\prod_j a_{\sigma(j)j},$$

where we substituted $j \mapsto \tau(j)$ in the product. Hence they cancel in pairs. ∎

Examples 6.2.4. For $n = 2$: $\det\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$. For $n = 3$ one obtains the *rule of Sarrus*:

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

There is *no* analogous "diagonal rule" for $n \geq 4$. If A is an upper (or lower) triangular matrix, only $\sigma = \mathrm{id}$ contributes, so $\det A = a_{11}a_{22}\cdots a_{nn}$.

6.3 Properties of the determinant

Theorem 6.3.1. *Let $A, B \in K^{n\times n}$.*

(a) $\det(AB) = \det A \cdot \det B$ *(multiplicativity)*

(b) A is invertible $\iff$ $\det A \neq 0$, and then $\det(A^{-1}) = (\det A)^{-1}$.

(c) $\det A^{\mathsf{T}} = \det A$. *In particular, all statements about columns hold for rows as well.*

(d) Similar matrices have the same determinant.

Proof. (a) Fix A and put $\delta(B) := \det(AB)$. Since the j-th column of AB is Ab_j, δ satisfies (D1) and (D2). By Theorem 6.2.3, $\delta(B) = \delta(I_n)\det B = \det A \det B$.

(b) If A is invertible, $1 = \det I_n = \det A \det A^{-1}$. If A is not invertible, its columns are dependent (Proposition 5.3.7), so $\det A = 0$ by Lemma 6.2.2(c).

(c) Since $\sigma \mapsto \sigma^{-1}$ is a bijection of S_n with $\operatorname{sgn}\sigma^{-1} = \operatorname{sgn}\sigma$, and $\prod_j a_{\sigma(j)j} = \prod_i a_{i\sigma^{-1}(i)}$ (reorder the factors by $i = \sigma(j)$),

$$\det A = \sum_{\sigma} \operatorname{sgn}(\sigma^{-1}) \prod_i a_{i\sigma^{-1}(i)} = \sum_{\rho} \operatorname{sgn}(\rho) \prod_i a_{i\rho(i)} = \det A^{\mathsf{T}}.$$

(d) $\det(S^{-1}AS) = \det(S)^{-1}\det A \det S = \det A$, using commutativity of K. ∎

Remark 6.3.2. By (d), the *determinant of an endomorphism* f of a finite-dimensional space, $\det f := \det M_{\mathcal{B}}^{\mathcal{B}}(f)$, is independent of the basis $\mathcal{B}$. Note that $\det(A+B) \neq \det A + \det B$ in general, and $\det(\lambda A) = \lambda^n \det A$.

Computing determinants in practice. Use row operations to reach a triangular matrix, keeping track of the effect: (R1) changes the sign, (R2) with factor λ multiplies the determinant by λ, (R3) does not change it. This needs $O(n^3)$ operations.

Example 6.3.3.

$$\det\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \end{pmatrix} = \det\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & -2 \\ 0 & 1 & 1 & 2 \\ 0 & -2 & 2 & 0 \end{pmatrix} = \det\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix} = -8.$$

Proposition 6.3.4 (Block triangular matrices)**.** *For $A \in K^{p\times p}$, $C \in K^{q\times q}$, $B \in K^{p\times q}$:* $\det\left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right) = \det A \cdot \det C$.

Proof. If $\det C = 0$, the last q rows of the block matrix are linearly dependent, so both sides vanish. Otherwise, for fixed B and invertible C, the map $A \mapsto \det\left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$ is multilinear and alternating in the columns of A, so by Theorem 6.2.3 it equals $\det\left(\begin{smallmatrix} I_p & B \\ 0 & C \end{smallmatrix}\right) \cdot \det A$. Clearing B by column operations (R3) using the first p columns gives $\det\left(\begin{smallmatrix} I_p & 0 \\ 0 & C \end{smallmatrix}\right) = \det C$, by the Leibniz formula. ∎

6.4 Laplace expansion and Cramer's rule

For $A \in K^{n\times n}$ let $A_{ij} \in K^{(n-1)\times(n-1)}$ denote the matrix obtained by deleting row i and column j.

Theorem 6.4.1 (Laplace expansion)**.** *For every $j \in [n]$ (expansion along column j) and every $i \in [n]$ (along row i):*

$$\det A = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det A_{ij} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det A_{ij}.$$

Proof. By (D1), $\det A = \sum_i a_{ij} \det A^{(i)}$, where $A^{(i)}$ is A with column j replaced by e_i. Moving row i to the top by $i-1$ adjacent row swaps and column j to the front by $j-1$ adjacent column swaps gives a matrix $\left(\begin{smallmatrix} 1 & * \\ 0 & A_{ij} \end{smallmatrix}\right)$, so $\det A^{(i)} = (-1)^{i+j} \det A_{ij}$ by Proposition 6.3.4. Row expansion follows by Theorem 6.3.1(c). ∎

Example 6.4.2 (Vandermonde determinant)**.** For $x_1, \dots, x_n \in K$,

$$V(x_1, \dots, x_n) := \det \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & & & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Proof by induction on n. Subtract x_1 times column k from column $k+1$, for $k = n-1, n-2, \dots, 1$ (in this order). Row i becomes $(1,\ x_i - x_1,\ x_i(x_i - x_1), \dots, x_i^{n-2}(x_i - x_1))$, and the first row becomes e_1^{T}. Expanding along the first row and pulling the factor $x_i - x_1$ out of row i for $i = 2, \dots, n$ gives $V(x_1, \dots, x_n) = \prod_{i=2}^{n}(x_i - x_1) \cdot V(x_2, \dots, x_n)$. ∎

In particular, $V \neq 0$ iff the x_i are pairwise distinct. Consequence: for pairwise distinct $x_1, \dots, x_n$ and arbitrary $y_1, \dots, y_n \in K$ there is exactly one polynomial $p \in K[x]_{\le n-1}$ with $p(x_i) = y_i$ for all i (*Lagrange interpolation*).

Definition 6.4.3. The *adjugate* of A is the matrix $\operatorname{adj}(A) \in K^{n\times n}$ with entries $\operatorname{adj}(A)_{ij} := (-1)^{i+j} \det A_{ji}$ (note the transposition of indices).

Theorem 6.4.4. $A \cdot \operatorname{adj}(A) = \operatorname{adj}(A) \cdot A = \det(A)\, I_n$. *Hence, if $\det A \neq 0$, then $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$.*

Proof. $(\operatorname{adj}(A)A)_{jk} = \sum_i (-1)^{i+j} a_{ik} \det A_{ij}$ is, by Theorem 6.4.1, the determinant of the matrix obtained from A by replacing column j with column k. This is $\det A$ if $j = k$ and 0 otherwise (two equal columns). The other product is treated using row expansion. ∎

Corollary 6.4.5 (Cramer's rule)**.** *If $A \in \mathrm{GL}_n(K)$ and $b \in K^n$, the unique solution of $Ax = b$ is given by $x_j = \det(a_1 | \cdots | a_{j-1} | b | a_{j+1} | \cdots | a_n) / \det A$.*

Proof. If $Ax = b$, then $b = \sum_k x_k a_k$; inserting this into column j and expanding by (D1) leaves only the term $x_j \det A$. ∎

Cramer's rule is of theoretical interest (e.g. it shows that the solution depends continuously – even rationally – on A and b), but it is useless for numerical computation.

Remark 6.4.6 (Geometric interpretation). For $K = \mathbb{R}$ and $n = 2$, $|\det(v|w)|$ is the area of the parallelogram spanned by v, w (see the cover picture), and $\det(v|w) > 0$ iff the rotation from v to w through the smaller angle is counter-clockwise. In general, $|\det A|$ is the factor by which f_A scales n-dimensional volumes – this is the origin of the Jacobian determinant in the transformation formula for integrals (Analysis III).

Chapter 7

Eigenvalues and Diagonalisation

Given an endomorphism f of a finite-dimensional space V, we ask for a basis in which the matrix of f is as simple as possible – ideally diagonal. Throughout, V is a K-vector space with $\dim V = n < \infty$.

7.1 Eigenvalues and eigenvectors

Definition 7.1.1. Let $f \in \mathrm{End}_K(V)$. A scalar $\lambda \in K$ is an *eigenvalue* of f if there is $v \in V$, $v \neq 0$, with $f(v) = \lambda v$; such a v is an *eigenvector* for λ. The *eigenspace* of λ is

$$\mathrm{Eig}(f, \lambda) := \ker(f - \lambda \,\mathrm{id}_V) = \{\, v \in V \mid f(v) = \lambda v \,\}.$$

f is *diagonalisable* if V has a basis consisting of eigenvectors of f. For $A \in K^{n\times n}$, these notions refer to f_A.

Thus λ is an eigenvalue iff $\mathrm{Eig}(f, \lambda) \neq \{0\}$, and the eigenvectors are the non-zero elements of the eigenspace. If $\mathcal{B} = (v_1, \ldots, v_n)$ is a basis of eigenvectors with $f(v_i) = \lambda_i v_i$, then $M_{\mathcal{B}}^{\mathcal{B}}(f) = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$; conversely, a diagonal matrix with respect to $\mathcal{B}$ means that $\mathcal{B}$ consists of eigenvectors. Hence, by Theorem 5.4.1:

Lemma 7.1.2. *$A \in K^{n\times n}$ is diagonalisable iff there is $S \in \mathrm{GL}_n(K)$ such that $S^{-1}AS$ is diagonal. The columns of such an S are then a basis of eigenvectors of A.*

Examples 7.1.3. (1) The reflection $f(x, y) = (x, -y)$ of $\mathbb{R}^2$ has eigenvalues 1 (eigenvector e_1) and -1 (eigenvector e_2).

(2) A rotation of $\mathbb{R}^2$ by an angle $\varphi \notin \pi\mathbb{Z}$ has no eigenvalue: no non-zero vector is mapped to a multiple of itself.

(3) $J = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has the only eigenvalue 1, and $\mathrm{Eig}(J, 1) = \mathrm{span}(e_1)$ is one-dimensional. So there is no basis of eigenvectors; J is not diagonalisable (over any field).

(4) On $C^\infty(\mathbb{R})$, every $\lambda \in \mathbb{R}$ is an eigenvalue of differentiation, with eigenvector $t \mapsto e^{\lambda t}$.

Theorem 7.1.4. *Let $v_1, \ldots, v_k$ be eigenvectors of f for pairwise distinct eigenvalues $\lambda_1, \ldots, \lambda_k$. Then $(v_1, \ldots, v_k)$ is linearly independent. More generally, the sum $\mathrm{Eig}(f, \lambda_1) + \cdots + \mathrm{Eig}(f, \lambda_k)$ is direct.*

Proof. Induction on k; $k = 1$ holds as $v_1 \neq 0$. Let $\sum_{i=1}^{k} \alpha_i v_i = 0$. Applying $f - \lambda_k \,\mathrm{id}$ yields $\sum_{i=1}^{k-1} \alpha_i(\lambda_i - \lambda_k)v_i = 0$. By induction hypothesis $\alpha_i(\lambda_i - \lambda_k) = 0$, hence $\alpha_i = 0$ for $i < k$, since $\lambda_i \neq \lambda_k$. Then $\alpha_k v_k = 0$ gives $\alpha_k = 0$. The second statement follows from the first and Lemma 4.5.2: if $u_1 + \cdots + u_k = 0$ with $u_i \in \mathrm{Eig}(f, \lambda_i)$ and some $u_i \neq 0$, then the non-zero u_i would be linearly dependent eigenvectors for distinct eigenvalues. ∎

Corollary 7.1.5. *f has at most n distinct eigenvalues. If it has n distinct eigenvalues, it is diagonalisable.*

7.2 The characteristic polynomial

λ is an eigenvalue of A iff $A - \lambda I_n$ is not invertible, iff $\det(A - \lambda I_n) = 0$. This motivates:

Definition 7.2.1. The *characteristic polynomial* of $A \in K^{n\times n}$ is

$$\chi_A := \det(xI_n - A) \in K[x],$$

where the determinant is computed with the Leibniz formula, the entries being polynomials. (The Leibniz formula makes sense over the commutative ring $K[x]$.)

Proposition 7.2.2. *(a)* $\chi_A = x^n - \operatorname{tr}(A)\,x^{n-1} + \cdots + (-1)^n \det A$, *where* $\operatorname{tr} A := a_{11} + \cdots + a_{nn}$ *is the* trace.

(b) Similar matrices have the same characteristic polynomial; so $\chi_f := \chi_{M_{\mathcal{B}}^{\mathcal{B}}(f)}$ *is well defined for* $f \in \operatorname{End}_K(V)$. *In particular* tr *and* det *are similarity invariants.*

(c) The eigenvalues of A *are exactly the roots of* χ_A *in* K.

Proof. (a) In the Leibniz formula for $\det(xI_n - A)$, every $\sigma \neq \mathrm{id}$ moves at least two indices, so the corresponding product has at most $n-2$ factors involving x. Hence the coefficients of x^n and x^{n-1} come from $\prod_i (x - a_{ii})$ alone. The constant term is $\chi_A(0) = \det(-A) = (-1)^n \det A$. (b) $xI - S^{-1}AS = S^{-1}(xI - A)S$, and the multiplicativity of det also holds for matrices over $K[x]$ (the proof of Theorem 6.3.1(a) can be repeated; alternatively, one argues with the field of rational functions). (c) For $\lambda \in K$, $\chi_A(\lambda) = \det(\lambda I - A)$, which vanishes iff λ is an eigenvalue. ∎

Example 7.2.3. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ we get $\chi_A = x^2 - (a+d)x + (ad - bc)$. The rotation matrix by φ has $\chi = x^2 - 2\cos\varphi\, x + 1$ with discriminant $-4\sin^2\varphi < 0$ for $\varphi \notin \pi\mathbb{Z}$: no real eigenvalues, but over $\mathbb{C}$ the eigenvalues $e^{\pm i\varphi}$. By Theorem 2.3.1, *every complex matrix has an eigenvalue.*

Definition 7.2.4. Let λ be an eigenvalue of f. Its *algebraic multiplicity* $\mu_a(f,\lambda)$ is the multiplicity of λ as a root of χ_f, i.e. the largest k with $(x-\lambda)^k \mid \chi_f$. Its *geometric multiplicity* is $\mu_g(f,\lambda) := \dim \operatorname{Eig}(f,\lambda)$.

Lemma 7.2.5. $1 \le \mu_g(f,\lambda) \le \mu_a(f,\lambda)$ *for every eigenvalue* λ.

Proof. Let $k = \mu_g(f,\lambda) \ge 1$ and extend a basis $(v_1,\dots,v_k)$ of $\operatorname{Eig}(f,\lambda)$ to a basis $\mathcal{B}$ of V. Then $M_{\mathcal{B}}^{\mathcal{B}}(f) = \left(\begin{smallmatrix} \lambda I_k & B \\ 0 & C \end{smallmatrix}\right)$, and by Proposition 6.3.4, $\chi_f = (x-\lambda)^k\,\chi_C$. ∎

7.3 Diagonalisability

Theorem 7.3.1 (Diagonalisability criterion)**.** *For* $f \in \operatorname{End}_K(V)$ *the following are equivalent:*

(i) f *is diagonalisable;*

(ii) $V = \operatorname{Eig}(f,\lambda_1) \oplus \cdots \oplus \operatorname{Eig}(f,\lambda_k)$, *where* $\lambda_1,\dots,\lambda_k$ *are the distinct eigenvalues of* f*;*

(iii) χ_f *splits into linear factors over* K, *and* $\mu_g(f,\lambda) = \mu_a(f,\lambda)$ *for every eigenvalue* λ.

Proof. Let $\lambda_1,\dots,\lambda_k$ be the distinct eigenvalues and $E_i = \operatorname{Eig}(f,\lambda_i)$, $d_i = \dim E_i$. By Theorem 7.1.4, $E := E_1 + \cdots + E_k$ is direct, so $\dim E = d_1 + \cdots + d_k$.

(i)⇒(ii): A basis of eigenvectors lies in E, so $E = V$. (ii)⇒(i): Concatenate bases of the E_i.

(ii)⇒(iii): Using the basis just described, $\chi_f = \prod_i (x-\lambda_i)^{d_i}$, so χ_f splits and $\mu_a(f,\lambda_i) = d_i$.

(iii)⇒(ii): If $\chi_f = \prod_i (x-\lambda_i)^{m_i}$ with $m_i = \mu_a(f,\lambda_i) = d_i$, then $\dim E = \sum d_i = \sum m_i = \deg \chi_f = n$, hence $E = V$. ∎

Algorithm. To diagonalise $A \in K^{n\times n}$: (1) compute χ_A and its roots; if it does not split, stop. (2) For each eigenvalue λ, find a basis of $\ker(A-\lambda I)$ by Gaussian elimination. (3) If the dimensions add up to n, the collected eigenvectors form the columns of S, and $S^{-1}AS$ is diagonal with the eigenvalues in the corresponding order.

Example 7.3.2. Let $A = \begin{pmatrix} 1 & 2 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix} \in \mathbb{R}^{3\times 3}$. As A is upper triangular, $\chi_A = (x-1)(x-3)(x-2)$, so A has three distinct eigenvalues and is diagonalisable. Solving $(A-\lambda I)v = 0$ gives

$$\lambda = 1:\ v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \lambda = 3:\ v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \lambda = 2:\ v_3 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

(Check: $Av_3 = (-2+2,\ -3+1,\ 2)^{\mathsf T} = (0,-2,2)^{\mathsf T} = 2v_3$.) With $S = (v_1|v_2|v_3)$ we get $S^{-1}AS = \operatorname{diag}(1,3,2)$.

Example 7.3.3. $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{smallmatrix}\right)$ has $\chi_A = (x-2)^2(x-3)$, but $\operatorname{Eig}(A,2) = \ker \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = \operatorname{span}(e_1)$ has dimension $1 < 2$. So A is not diagonalisable, although χ_A splits.

7.4 An application: linear recursions

Diagonalisation makes powers easy: if $A = SDS^{-1}$ with $D = \operatorname{diag}(\lambda_1,\dots,\lambda_n)$, then $A^k = SD^kS^{-1}$ with $D^k = \operatorname{diag}(\lambda_1^k,\dots,\lambda_n^k)$.

Example 7.4.1 (Fibonacci numbers)**.** Let $F_0 = 0$, $F_1 = 1$, $F_{k+1} = F_k + F_{k-1}$. With $u_k = (F_{k+1}, F_k)^{\mathsf T}$ we have $u_k = Au_{k-1}$ for $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so $u_k = A^k u_0$. Now $\chi_A = x^2 - x - 1$ has the roots $\varphi = \frac{1+\sqrt5}{2}$ and $\psi = \frac{1-\sqrt5}{2}$, with eigenvectors $(\varphi,1)^{\mathsf T}$ and $(\psi,1)^{\mathsf T}$. Writing $u_0 = (1,0)^{\mathsf T} = \frac{1}{\sqrt5}\big((\varphi,1)^{\mathsf T} - (\psi,1)^{\mathsf T}\big)$ we obtain $u_k = \frac{1}{\sqrt5}\big(\varphi^k(\varphi,1)^{\mathsf T} - \psi^k(\psi,1)^{\mathsf T}\big)$, and reading off the second component,

$$F_k = \frac{1}{\sqrt5}\left(\Big(\frac{1+\sqrt5}{2}\Big)^k - \Big(\frac{1-\sqrt5}{2}\Big)^k\right) \qquad \text{(formula of Binet).}$$

Since $|\psi| < 1$, F_k is the integer closest to $\varphi^k/\sqrt5$.

7.5 Outlook

Not every endomorphism is diagonalisable, even over $\mathbb{C}$ (Examples 7.1.3(3)). In *Linear Algebra II* we shall prove:

- *Triangularisation:* if χ_f splits, there is a basis in which f has upper triangular form.
- *Cayley–Hamilton:* $\chi_f(f) = 0$ for every endomorphism f. (For 2×2 matrices this can be verified by a direct computation, cf. Exercise 24.)
- *Jordan normal form:* if χ_f splits, f has a matrix that is block diagonal with blocks of the form $\lambda I + N$, N having ones on the superdiagonal – this completely classifies matrices over $\mathbb{C}$ up to similarity.
- *Spectral theorem:* every real symmetric matrix is diagonalisable, even with an *orthonormal* basis of eigenvectors.

Supplement: invariant subspaces

A subspace $U \subseteq V$ is *f-invariant* if $f(U) \subseteq U$. Eigenspaces, $\ker f$ and $\operatorname{im} f$ are f-invariant, and a vector $v \neq 0$ is an eigenvector iff the line $\operatorname{span}(v)$ is invariant. If U is invariant and $(u_1, \ldots, u_k)$ is a basis of U, extended to a basis $\mathcal{B}$ of V, then

$$M_{\mathcal{B}}^{\mathcal{B}}(f) = \begin{pmatrix} A_1 & B \\ 0 & C \end{pmatrix},$$

where $A_1 \in K^{k \times k}$ is the matrix of $f|_U$ with respect to $(u_1, \ldots, u_k)$. Hence $\chi_{f|_U}$ divides χ_f by Proposition 6.3.4. If $V = U \oplus W$ with *both* U and W invariant, then $B = 0$ and the matrix is block diagonal. Decomposing V into a direct sum of invariant subspaces that are as small as possible is the guiding principle of the normal form theory in Linear Algebra II; diagonalisability is precisely the case in which V decomposes into invariant *lines*.